A Structural Characterisation of the 3×3 Magic Square of Squares

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Abstract

We give a complete parametrisation of the family of 3×3 magic squares, expressed so as to isolate the arithmetic difficulty in the problem of nine distinct perfect squares. Writing the nine cells in terms of three free parameters, we show that the linear system defining the family has rank 6 and nullity 3, and we exhibit an explicit basis for its solution space. We identify a distinguished triple of *anchors* (E^2, I^2, X) in terms of which three of the cells take a symmetric form, and show that the resulting configuration is the medial triangle of the cells it determines. Normalising by scale and translation reduces the configuration to a one-parameter family. We then locate the obstruction to a nine-square solution precisely: it is the requirement that $2E^2$ admit four essentially distinct representations as a sum of two squares, subject to one additional linear condition. Finally we prove that no linear relation among the cells can obstruct a nine-square solution, which explains why purely algebraic approaches to the problem do not close.

1 Setup

Label the 3×3 square

A^2	B^2	C^2
D^2	E^2	F^2
X	H^2	I^2

where E^2 denotes the centre and X the bottom-left cell. The notation records our intended application — that the entries be perfect squares — but no such assumption is made in Sections 2–5, where all results hold for arbitrary real entries satisfying the magic condition.

2 The Parametrisation

Proposition 1. *Every 3×3 magic square is determined by A^2, E^2 and X through*

$$\begin{aligned}
C^2 &= 2E^2 - X, && (1)\\
I^2 &= 2E^2 - A^2, && (2)\\
B^2 &= E^2 - A^2 + X, && (3)\\
H^2 &= E^2 + A^2 - X, && (4)\\
D^2 &= 3E^2 - A^2 - X, && (5)\\
F^2 &= A^2 + X - E^2, && (6)
\end{aligned}$$

and its magic sum is $3E^2$.

Proof. The eight line conditions are linear in the nine cells. Solving the diagonal, the first column and the first row successively for I^2, X-complement and C^2 yields (1)–(6); each remaining line is then satisfied identically. □

Corollary 2 (Opposite pairs)**.** *Each pair of cells symmetric about the centre averages to it:*

$$A^2 + I^2 = B^2 + H^2 = C^2 + X = D^2 + F^2 = 2E^2.$$

Consequently every line consists of the centre together with one opposite pair, and sums to $E^2 + 2E^2 = 3E^2$.

3 Relations Satisfied by B^2

Since B^2 is determined by the parameters, it may be written in terms of each of the remaining cells:

$$\begin{aligned}
B^2 &= E^2 - A^2 + X, & B^2 &= 3E^2 - A^2 - C^2,\\
B^2 &= E^2 - C^2 + I^2, & B^2 &= 2I^2 - D^2,\\
B^2 &= 2E^2 - 2A^2 + F^2, & B^2 &= -A^2 + F^2 + I^2,\\
B^2 &= 2X - F^2, & B^2 &= 2E^2 - H^2.
\end{aligned}$$

As a coefficient matrix over the ordered basis $(E^2, A^2, C^2, D^2, F^2, X, H^2, I^2)$ these read

$$M = \begin{pmatrix}
1 & -1 & 0 & 0 & 0 & 1 & 0 & 0\\
3 & -1 & -1 & 0 & 0 & 0 & 0 & 0\\
1 & 0 & -1 & 0 & 0 & 0 & 0 & 1\\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 2\\
2 & -2 & 0 & 0 & 1 & 0 & 0 & 0\\
0 & -1 & 0 & 0 & 1 & 0 & 0 & 1\\
0 & 0 & 0 & 0 & -1 & 2 & 0 & 0\\
2 & 0 & 0 & 0 & 0 & 0 & -1 & 0
\end{pmatrix}.$$

Lemma 3. *Every row of M has coefficient sum 1.*

Proof. The constant square, with all nine entries equal to t, is magic. Substituting it into any valid relation gives $t = \left(\sum_j m_{ij}\right)t$ for all t. $\square$

The row sums therefore serve as a consistency check on any derived relation. Moreover $\det M = 0$: for instance $r_5 + r_7 = 2r_1$, since $(2E^2 - 2A^2 + F^2) + (2X - F^2) = 2(E^2 - A^2 + X)$.

Proposition 4. *The system* (1)–(6) *has rank* 6 *and nullity* 3. *Every linear relation holding on the family is a linear combination of these six, and a basis for the solution space, in coordinates* $(E^2, A^2, C^2, D^2, F^2, X, H^2, I^2; B^2)$, *is*

$$\begin{aligned}
v_1 &= (1, 0, 2, 3, -1, 0, 1, 2;\ 1),\\
v_2 &= (0, 1, 0, -1, 1, 0, 1, -1;\ -1),\\
v_3 &= (0, 0, -1, -1, 1, 1, -1, 0;\ 1).
\end{aligned}$$

Remark 5. Any further relation derived from the system — however many substitutions it involves, and whatever constant multiple of E^2 it closes at — is by Proposition 4 a combination of the six. It is therefore an identity on the family, valid for every magic square, and imposes no condition.

4 The Anchors (E^2, I^2, X)

By Corollary 2, each of E^2, I^2, X is the mean of a pair drawn from $\{B^2, D^2, F^2\}$:

$$D^2 + F^2 = 2E^2, \qquad B^2 + D^2 = 2I^2, \qquad B^2 + F^2 = 2X.$$

Solving for the cells gives the symmetric form

$$\begin{aligned}
B^2 &= -E^2 + I^2 + X, && (7)\\
D^2 &= E^2 + I^2 - X, && (8)\\
F^2 &= E^2 - I^2 + X, && (9)
\end{aligned}$$

the three expressions differing only in the placement of the minus sign. The remaining cells follow as $A^2 = 2E^2 - I^2$, $C^2 = 2E^2 - X$ and $H^2 = 2E^2 - B^2$, so (E^2, I^2, X) may be taken as free parameters in place of (A^2, E^2, X).

Proposition 6 (Medial configuration)**.** *If B^2, D^2, F^2 are taken as the vertices of a triangle, then I^2, X, E^2 are the midpoints of the edges B^2D^2, B^2F^2 and D^2F^2 respectively. Each cell is the sum of the two anchors incident to it, minus the anchor on the opposite edge.*

Differencing the identities eliminates the shared cell and doubles the remaining anchors:

$$B^2 - D^2 = 2(X - E^2), \quad B^2 - F^2 = 2(I^2 - E^2), \quad D^2 - F^2 = 2(I^2 - X).$$

Thus differences of cells are twice the corresponding differences of anchors, and each anchor comparison mirrors a cell comparison: for example $I^2 > X$ precisely when $D^2 > F^2$. The relation $B^2 - F^2 = 2(I^2 - E^2)$ is the midline theorem for the triangle above.

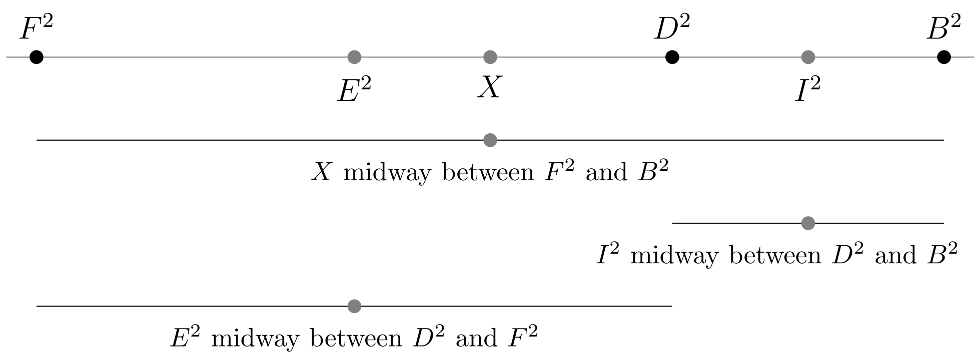
5 Normalisation

The configuration has three degrees of freedom; removing scale and translation leaves one. Place F^2 at the origin and B^2 at 1, and set

$$t = \frac{D^2 - F^2}{B^2 - F^2}.$$

Point	Position	Origin
F^2	0	endpoint
E^2	$t/2$	midpoint of D^2, F^2
X	$1/2$	midpoint of F^2, B^2
D^2	t	free parameter
I^2	$(1+t)/2$	midpoint of D^2, B^2
B^2	1	endpoint

Two features are independent of t: the anchor X is fixed at $1/2$, and $I^2 - E^2 = 1/2$, so E^2 and I^2 remain a fixed half-unit apart as t varies. The remaining gaps are $I^2 - X = t/2$ and $X - E^2 = (1-t)/2$. The values $t = 0$ and $t = 1$ are excluded, as they force $D^2 = F^2$ and $D^2 = B^2$ respectively.



The figure is drawn to scale for the square of Section 8, for which $t \approx 0.7006$. The midpoint structure is universal; the left-to-right ordering is not, and permutes with t.

De-normalised form. Writing $S = B^2 - F^2$ for the scale, the table above rescales to

$$X = F^2 + \tfrac{1}{2}S, \qquad E^2 = F^2 + \tfrac{t}{2}S, \qquad D^2 = F^2 + tS, \qquad I^2 = F^2 + \tfrac{1+t}{2}S.$$

Each cell is t — a fixed shape, invariant under scaling — combined with the two free parameters F^2 and S, which carry the square's actual size. For the square of Section 8, $F^2 = 83521$ and $S = 277200$, and these reproduce $X = 222121$, $E^2 = 180625$, $D^2 = 277729$, $I^2 = 319225$ exactly.

6 A Second Normalisation: D^2 and B^2

The Medial Configuration has three vertices, B^2, D^2, F^2, and any two of them may be placed at the endpoints, with the third left as the free parameter. Section 5 fixed F^2 and B^2; fixing D^2 and B^2 instead, and setting

$$t' = \frac{F^2 - D^2}{B^2 - D^2},$$

gives the same six points a second, equally valid labelling.

Point	Position	Origin
D^2	0	endpoint
E^2	$t'/2$	midpoint of D^2, F^2
X	$(1+t')/2$	midpoint of F^2, B^2
F^2	t'	free parameter
I^2	$1/2$	midpoint of D^2, B^2
B^2	1	endpoint

The roles have swapped: it is now I^2 that sits fixed at $1/2$, and $X - E^2 = 1/2$ that stays constant as t' varies, mirroring how X and $I^2 - E^2$ were the fixed quantities in Section 5. As before, $t' = 0$ and $t' = 1$ are excluded, forcing $F^2 = D^2$ and $F^2 = B^2$ respectively.

For the square of Section 8, $t' = -578/247 \approx -2.340$ — negative, and outside $[0, 1]$, because F^2 is the smallest of the three vertices B^2, D^2, F^2, not a point strictly between D^2 and B^2. The six points themselves are unchanged — this is the same diagram as Section 5, read with a different pair fixed as 0 and 1 — only their coordinates differ: $D^2 = 8.407$, $I^2 = 10.204$, $B^2 = 12$ in the earlier figure's units, with F^2 now falling to the left of the origin rather than at it.

De-normalised form. Writing $S' = B^2 - D^2$,

$$X = D^2 + \tfrac{1}{2}S', \qquad E^2 = D^2 + \tfrac{t'}{2}S', \qquad F^2 = D^2 + t'S', \qquad I^2 = D^2 + \tfrac{1}{2}S'.$$

For the square of Section 8, $D^2 = 277729$ and $S' = 82992$, reproducing $F^2 = 83521$, $E^2 = 180625$, $X = 222121$, $I^2 = 319225$ exactly.

7 The Determinant

Proposition 7. *For any 3×3 magic square in the above notation,*

$$\det = 9E^2\,(F^2 - E^2)(X - A^2) = 9E^2\big[(X - E^2)^2 - (A^2 - E^2)^2\big].$$

Since $9E^2 = (3E)^2$, the determinant is a perfect square exactly when $(F^2 - E^2)(X - A^2)$ is a non-negative perfect square. It vanishes precisely when $F^2 = E^2$ or $X = A^2$, both degenerate cases with a repeated entry.

8 Example

The square of Bremner and Sallows, with centre 425^2, contains seven perfect squares:

205^2	360721	373^2
527^2	425^2	289^2
222121	23^2	565^2

Here $A^2 = 42025$, $E^2 = 180625$ and $X = 222121$, and the parametrisation returns every remaining cell. The four opposite pairs each sum to $2E^2 = 361250$, and every line sums to $3E^2 = 541875$. The two non-square entries occupy the cells B^2 and X, which we accordingly call the *closing cells*: once the other seven entries are fixed, these two are determined, and whether they are squares is not settled by the linear structure.

9 Where the Obstruction Lies

Write the entries as $a^2, \ldots, i^2$ with $X = g^2$. By Corollary 2 a nine-square magic square requires

$$a^2 + i^2 = b^2 + h^2 = c^2 + g^2 = d^2 + f^2 = 2e^2,$$

that is, four essentially distinct representations of $2e^2$ as a sum of two squares, together with the single linear condition $a^2+b^2+c^2 = 3e^2$. The number of such representations is governed by the primes $p \equiv 1 \pmod 4$ dividing e, so any candidate centre must be divisible by several of them.

Equivalently, in the anchor form (7)–(9), the five cells B^2, D^2, F^2, I^2, X constitute two three-term arithmetic progressions of squares sharing the endpoint B^2. Each progression is individually easy to realise; it is their simultaneous closure, together with the corresponding conditions on A^2, C^2 and H^2, that has not been achieved.

Theorem 8. *No linear relation among the nine cells can be violated by requiring that the entries be perfect squares.*

Proof. By Proposition 4 every linear relation holding on the family is a combination of (1)–(6), hence holds identically at every point of the three-dimensional solution space. A nine-square magic square, if one exists, is such a point, and therefore satisfies every such relation. □

Theorem 8 accounts for a persistent feature of algebraic attacks on this problem. Successive elimination among the defining equations produces relations closing at $11E^2$, $9E^2$, $7E^2$, $5E^2$, $3E^2$ and E^2, and each is correct; but each is an identity, and none excludes any configuration. Any resolution must therefore invoke the arithmetic of representations by sums of two squares, and not the linear structure alone.

10 An Illustrative Identity

The following relation, obtained by successive elimination, involves every cell of the square:

$$X = 11E^2 - X - D^2 - 2H^2 - 2I^2 - (A^2 + B^2 + F^2 + C^2). \tag{II1}$$

Regrouping so that each cell stands beside the multiple of its opposite partner exhibits the balance:

$$X = 11E^2 - X - \underbrace{(D^2 + F^2)}_{2E^2} - \underbrace{(B^2 + 2H^2)}_{2E^2+H^2} - \underbrace{(A^2 + 2I^2)}_{2E^2+I^2} - C^2.$$

Each brace is an instance of Corollary 2: for example $A^2 + 2I^2 = (A^2 + I^2) + I^2 = 2E^2 + I^2$. Substituting all three reduces the constant from $11E^2$ to $5E^2$,

$$X = 5E^2 - X - H^2 - I^2 - C^2,$$

and applying $C^2 = 2E^2 - X$ once more gives

$$2X = 6E^2 - 2H^2 - 2I^2, \qquad \text{that is} \qquad X = 3E^2 - H^2 - I^2,$$

which is the bottom row of the square.

Continuing the same elimination from (II1) produces a chain of relations closing successively at $9E^2$, $7E^2$, $5E^2$, $3E^2$ and finally E^2; each is valid, and each reduces to a line of the square already recorded in Section 2. The constant at which a relation closes therefore records the number of substitutions applied in reaching it, and is not an invariant of the square. By Proposition 4 no such relation can be otherwise: the system has rank 6, so every derived relation is a combination of (1)–(6) and holds identically on the family.

For the square of Section 8, (II1) evaluates to $1986875-222121-277729-1058-638450-625396 = 222121 = X$, as required.

Continuing the same elimination from (II1) produces a chain of relations, each obtained from the previous one by trading a single opposite pair for $2E^2$:

$$X = 11E^2 - X - D^2 - 2H^2 - 2I^2 - (A^2 + B^2 + F^2 + C^2), \tag{II1}$$
$$X = 9E^2 - 2H^2 - 2I^2 - (A^2 + D^2 + F^2 + B^2), \tag{II2}$$
$$X = 7E^2 - 2I^2 - (A^2 + D^2 + F^2 + H^2), \tag{II3}$$
$$X = 5E^2 - (I^2 + D^2 + F^2 + H^2), \tag{II4}$$
$$X = 3E^2 - (I^2 + H^2), \tag{II5}$$
$$X = \ E^2 + B^2 - I^2. \tag{II6}$$

Each substitution removes two cells and lowers the constant by 2, so the chain descends $11, 9, 7, 5, 3, 1$ and terminates once every opposite pair has been consumed. All six relations evaluate to $X = 222121$ on the square of Section 8.

The endpoint is not arbitrary. Rearranging (II6) gives

$$B^2 = -E^2 + I^2 + X,$$

which is precisely the anchor form (7). The chain therefore closes at E^2 because only the three anchors E^2, I^2, X remain, and these are a minimal parameter set for the family: no further substitution is available. The elimination chain and the anchor form are the same statement, reached from opposite ends.

Consequently the constant at which a relation closes records the number of substitutions applied in reaching it, and is not an invariant of the square. By Proposition 4 no such relation can be otherwise: the system has rank 6, so every derived relation is a combination of (1)–(6) and holds identically on the family.

11 Nested Radical Chains for the Other Cells

The elimination chain (II1)–(II6) of Section 10 is not particular to X. The same phenomenon — an arbitrarily long derivation that nonetheless collapses to one of the six defining relations (1)–(6) — occurs for every cell, and is conveniently generated by repeated application of the partner function g of Definition 48. Four such chains, computed for the square of Section 8 ($A = 205$, $D = 527$, $E = 425$), illustrate the point. Write $X = 3E^2 - A^2 - D^2$ and $C^2 = 2E^2 - X$ as before.

Nesting g four layers deep reaches H:

$$H = \sqrt{2E^2 - \Big(3E^2 - A^2 - \big(2E^2 - (3E^2 - A^2 - D^2)\big)\Big)} = 23,$$

where the innermost layer is X, the next is C^2, and the layer before the final root is B^2. A shorter route reaches F from I^2 and C^2 directly, bypassing B^2 and H^2 entirely:

$$F = \sqrt{\big(3E^2 - I^2\big) - C^2} = 289, \qquad I^2 = 2E^2 - A^2.$$

Reusing the same H from the first chain, together with the column sum, closes a loop back to I:

$$I = \sqrt{\big(A^2 + D^2\big) - H^2} = 565.$$

A fourth, still longer chain reaches F a second way, replacing I^2 above with its own chain form $\big(A^2 + D^2\big) - H^2$ rather than the shortcut $2E^2 - A^2$:

$$F = \sqrt{3E^2 - C^2 - I^2} = 289.$$

Written out in full, with every intermediate cell — X, C^2, B^2, H^2, and finally I^2 — expanded back to A^2, D^2, and E^2 alone, this is a single unbroken nested radical seven layers deep:

$$F = \big(3E^2 - \big(2E^2 - \big(3E^2 - A^2 - D^2\big)\big) - \big(3E^2 - \big(3E^2 - A^2 - D^2\big) - \big(2E^2 - \big(3E^2 - A^2 - \big(2E^2 - \big(3E^2 - A^2 - D^2\big)\big)\big)\big)\big)\big)^{1/2} = 289$$

Reading from the innermost parentheses outward recovers the whole chain in one expression: $X = 3E^2 - A^2 - D^2$ appears three times over (once for C^2, once directly inside the I^2 term, and once again inside the H^2 buried within it); the innermost application of $2E^2 - X$ gives C^2; wrapping that in $3E^2 - A^2 - (\cdot)$ and then $2E^2 - (\cdot)$ gives B^2 and then H^2; and the two remaining outer layers assemble $I^2 = (A^2 + D^2) - H^2$ and finally $F^2 = 3E^2 - C^2 - I^2$. Every layer is one of the six equations (1)–(6) applied in turn, exactly as Proposition 9 requires.

Proposition 9. *Every nested-radical expression built from E^2, A^2, D^2 and repeated applications of* (1)–(6) *collapses, once simplified, to one of the six defining equations.*

Proof. Immediate from Proposition 4: the system has rank 6, so every relation derivable from it, however many substitutions are used to reach it, is a linear combination of (1)–(6) and holds identically on the family. □

The fourth chain illustrates this concretely: expanding $3E^2 - C^2 - I^2$ and using $C^2 + I^2 = D^2 + E^2$ (Section 38) gives $3E^2 - C^2 - I^2 = 3E^2 - (D^2 + E^2) = 2E^2 - D^2$, which is (6) itself. The extra layers of nesting record only how many substitutions were taken to reach that one relation, exactly as in the chain (II1)–(II6).

Cell	Value at $A = 205$, $D = 527$, $E = 425$
H	23
F	289
I	565
F	289 (second, longer chain)

12 The Chain as a Walk Around the Square

The nested radical for F in Section 11 alternates between $2E^2$ and $3E^2$ at successive layers. This is not an arbitrary feature of the algebra: each layer is a distinct geometric identity of the square — a line (three cells, summing to $3E^2$) or an opposite pair (two cells, summing to $2E^2$) — and the chain, read from the innermost layer outward, is a literal walk around the grid.

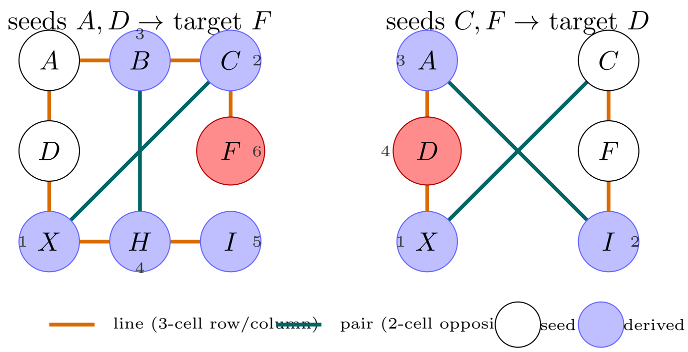
Layer	Identity used	Multiple of E^2
X	column 1: $A^2 + D^2 + X = 3E^2$	$3E^2$
C^2	corner pair: $C^2 + X = 2E^2$	$2E^2$
B^2	top row: $A^2 + B^2 + C^2 = 3E^2$	$3E^2$
H^2	edge pair: $B^2 + H^2 = 2E^2$	$2E^2$
I^2	bottom row: $X + H^2 + I^2 = 3E^2$	$3E^2$
F^2	right column: $C^2 + F^2 + I^2 = 3E^2$	$3E^2$

Each row of the table is a genuine geometric line or pair of the square: $A^2+D^2+X = 3E^2$ is column 1; $C^2 + X = 2E^2$ is Corollary 2's corner pair; $A^2 + B^2 + C^2 = 3E^2$ is the top row; $B^2+H^2 = 2E^2$ is the edge pair; $X+H^2+I^2 = 3E^2$ is the bottom row; and $C^2+F^2+I^2 = 3E^2$ is the right column.

The first five layers alternate perfectly — line, pair, line, pair, line — tracing a path down the left column to X, across to the corner pair for C, along the top row for B, across to the edge pair for H, and along the bottom row for I. The alternation breaks only at the sixth and final layer: the one remaining edge pair, (D, F), would continue it, but D is already known (it is a seed), so that pair would supply only the trivial shortcut $F^2 = 2E^2 - D^2$, not a new derivation. The chain instead reaches F through the one remaining *line*, the right column $C^2 + F^2 + I^2 = 3E^2$, giving two line-layers in succession at the close.

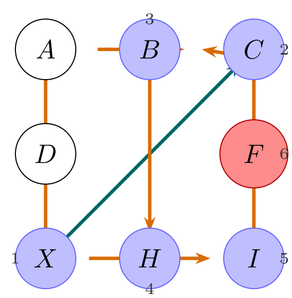
Remark 10. Of the square's eight defining identities — four lines and four pairs — the chain uses six: three lines (column 1, top row, bottom row) and two pairs (the corner pair (C, X) and the edge pair (B, H)), leaving the other corner pair (A, I) and the other edge pair (D, F) unused as independent derivations, since A and D are already fixed as seeds and I, once known, only offers the immediately-checkable identity $A^2 + I^2 = 2E^2$ rather than a new quantity. The nested radical is therefore not merely a long formula: it is a record of which six of the square's eight lines and pairs were traversed, and in what order, to reach F from the two seeds A and D.

Drawing the walk directly on the grid makes the geometry explicit. A second, shorter walk is available from the seeds C, F: the pair (C, X) gives X; column 3 ($C^2+F^2+I^2 = 3E^2$) gives I^2 directly from the two seeds; the pair (A, I) then gives A^2; and column 1 closes at D^2 — four steps, alternating pair, line, pair, line, and never touching B or H at all.



13 Recurrence Across Chains: the Role of X

Section 11 built one long chain from A, D, E, reaching F through X, C^2, B^2, H^2, I^2. Drawn as a dependency graph on the grid, every step of that chain is a single clean geometric identity — none is a combination of two — which is why the alternation of Section 12 is so regular.



A second, independent chain, seeded instead by C, F, E, reaches D through the same kind of nesting. Solving the six defining equations for A^2 and X in terms of C^2, F^2, E^2 gives $X = 2E^2 - C^2$ and $A^2 = C^2 + F^2 - E^2$; chaining forward from there, and using

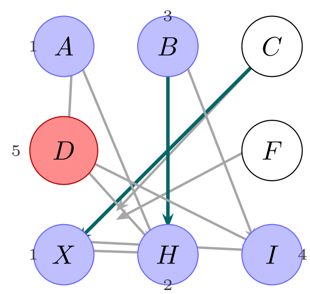
$$H^2 + I^2 = (E^2 + A^2 - X) + (2E^2 - A^2) = 3E^2 - X = A^2 + (3E^2 - A^2 - X) = A^2 + D^2,$$

— itself a direct instance of the redundancy in Proposition 4 — gives $D^2 = H^2 + I^2 - A^2$ and hence:

$$D = \left((E^2 + (C^2 + F^2 - E^2) - (2E^2 - C^2)) + (E^2 + (2E^2 - (E^2 + (C^2 + F^2 - E^2) - (2E^2 - C^2))) - (2E^2 - C^2)) - (C^2 + F^2 - E^2)\right)^{1/2} = 527$$

Reading it out: $(C^2 + F^2 - E^2)$ is A^2, $(2E^2 - C^2)$ is X, the first large parenthesis is H^2, the second is I^2, and $H^2 + I^2 - A^2$ closes at D^2.

Drawn the same way, this chain looks markedly different: most of its steps combine two prior identities rather than applying one, since Proposition 51 itself already draws on two of the square's identities before this chain reuses it further.



Only $C \to X$ and $H \to B$ (teal) are single clean pair identities; every other arrow (gray) combines two prior quantities at once, and the final step merges three (H^2, I^2, and A^2) to close at D^2. Eleven dependency edges in all, against the first chain's ten — but concentrated into five tiers instead of six, and mostly combined rather than clean, which is exactly why the shorter, all-clean four-step route of Section 12 is the better path to D.

Comparing the two chains lets a question be asked precisely: which cell recurs the most as a *derived* quantity, once seeds — which cost nothing to reference, since they are simply given — are excluded from the count?

Chain	Seeds	X occurrences	Next-most-recurrent
$A, D, E \to F$	A, D, E	3	C^2: 2
$C, F, E \to D$	C, F, E	3	A^2: 3 (tied)

In the first chain X recurs more than any other derived quantity; in the second it ties with A^2, which — unlike in the first chain, where A is a free seed — must itself be computed from C, F, E before it can be reused at all. The precise statement is not that X is the unique recurring cell, but that it is never surpassed: across two independent choices of seed pair, X recurs at least as often as anything else derived, tying at worst. This follows from Proposition 4: $\{A^2, E^2, X\}$ is the three-parameter basis underlying all six defining equations (1)–(6), so X — together with whichever cell plays A^2's role in a given chain — is structurally load-bearing for every other cell, appearing in five of the six defining relations.

This structural centrality is distinct from, though complementary to, a second and separate asymmetry recorded in Section 38: swapping which of two representation-list values is called A and which D changes the outcome, since $H^2 = E^2 + A^2 - X$ treats A and D unequally — $(A, D) = (205, 527)$ reaches seven squares, while $(A, D) = (527, 205)$ reaches only six. That asymmetry is geometric, arising from A occupying the top-left corner (shared by the top row, the main diagonal, and the first column) while D occupies only the middle-left edge (shared by the middle row and the first column); it is unrelated to X's algebraic centrality, and neither fact explains the other.

Remark 11. It is fair, in this light, to speak informally of A as the cell whose *position* forces an asymmetric start — which of two candidate values is placed there changes the count — and of X as the cell that recurs throughout the computation, never less than any other derived quantity. Both are true statements about this square, arrived at independently; neither is the cause of the other. In particular, the outcome that C turns out square while X does not (Section 39) is settled by X's value the moment it is first computed — from C^2 alone, or from $A^2 + D^2$, whichever chain one is following — and not by anything that happens to X later in a chain that goes on to reuse it.

14 A Catalog of Chain Shapes

Sections 11, 12, and 13 each built a nested-radical chain reaching one target cell from one seed pair. Collected together, in terms of E^2 and the two seeds alone, the chains found for the square of Section 8 are:

Seeded by A, D, E

$$H = \sqrt{2E^2 - \Big(3E^2 - A^2 - \big(2E^2 - (3E^2 - A^2 - D^2)\big)\Big)} = 23.$$

$$I = \sqrt{(A^2 + D^2) - \Big(2E^2 - \big(3E^2 - A^2 - (2E^2 - (3E^2 - A^2 - D^2))\big)\Big)} = 565.$$

$$F = \sqrt{\big(3E^2 - (2E^2 - A^2)\big) - \big(2E^2 - (3E^2 - A^2 - D^2)\big)} = 289.$$

The full seven-layer chain to F that instead routes through C^2 and I^2 both fully expanded is given in Section 11.

Seeded by C, F, E

$$X = 2E^2 - C^2 = 222121, \qquad I = \sqrt{3E^2 - C^2 - F^2} = 565,$$
$$A = \sqrt{2E^2 - (3E^2 - C^2 - F^2)} = 205.$$

$$D = \sqrt{3E^2 - \big(2E^2 - (3E^2 - C^2 - F^2)\big) - (2E^2 - C^2)} = 527$$

$$B = \sqrt{2E^2 - \Big(E^2 + (C^2 + F^2 - E^2) - (2E^2 - C^2)\Big)} = 600.6005328002965\ldots$$

$$I = \sqrt{E^2 + \Big(2E^2 - \big(E^2 + (C^2 + F^2 - E^2) - (2E^2 - C^2)\big)\Big) - (2E^2 - C^2)} = 565$$

The four-step chain above for D is the clean route of Section 12; the six-layer chain of Section 13, which instead passes through H^2 and B^2, reaches the same D by a longer, mostly-combined path. The last line recovers I a second way, reusing the B-chain rather than the direct pair identity $I^2 = 2E^2 - A^2$ — the same value, 565, by three independent routes now on record: directly from A^2, via the A, D, E chain above, and via this B-chain.

Seeds	Target	Layers	Note
A, D, E	H	4	shortest route to H
A, D, E	I	5	reuses the H-chain
A, D, E	F	3	shortest route to F
A, D, E	F	7	Section 11, fully expanded
C, F, E	X, I, A	1–2	direct, no chaining needed
C, F, E	D	4	Section 12, all-clean
C, F, E	D	6	Section 13, mixed
C, F, E	B	4	
C, F, E	I	5	third route to I

Remark 12. Every entry in this catalog is guaranteed, by Proposition 9, to reduce to one of the six defining equations (1)–(6) however long it is written — the table records how many layers of substitution a particular choice of route happens to take, not a fact about the cell reached. That several distinct chains reach the same value (565 for I, by three separate routes above) is exactly what rank 6 predicts: with only three free parameters, most quantities of interest admit more than one path to the same answer.

The remaining seed pairs

Twelve pairs of outer cells are valid seeds: any two cells sharing a row or column, excluding the four opposite pairs $(A, I), (B, H), (C, X), (D, F)$, which are never independent. They fall into four families, one per outer line, and $A, D, E \to F$ (Sections 11, 12) and $C, F, E \to D$ (Section 12) already cover two of the four. A deep, single-thread walk exists for each of the remaining two families as well.

Row 1: (A, B), (A, C), (B, C). Represented by (A, C), reaching F:

$$B^2 = 3E^2 - A^2 - C^2, \quad I^2 = 2E^2 - A^2, \quad H^2 = 2E^2 - B^2,$$

$$X = 2E^2 - C^2, \quad D^2 = 3E^2 - A^2 - X,$$

$$F^2 = 3E^2 - C^2 - I^2 = 83521, \qquad F = 289.$$

(A, B) and (B, C) reach the same six values by the same identities, after one preliminary step to recover the missing row-1 member: $C^2 = 3E^2 - A^2 - B^2$ for (A, B), or $A^2 = 3E^2 - B^2 - C^2$ for (B, C).

Row 3: (X, H), (X, I), (H, I). Represented by (H, I), reaching F:

$$X = 3E^2 - H^2 - I^2, \quad A^2 = 2E^2 - I^2, \quad B^2 = 2E^2 - H^2, \quad C^2 = 2E^2 - X, \quad D^2 = 3E^2 - A^2 - X,$$

$$F^2 = 3E^2 - C^2 - I^2 = 83521, \qquad F = 289.$$

(X, H) and (X, I) reach the same six values after recovering the missing row-3 member: $I^2 = 3E^2 - X - H^2$ for (X, H), or $H^2 = 3E^2 - X - I^2$ for (X, I).

Column 1: (A, D), (A, X), (D, X). (A, D) is Sections 11–12. Represented instead by (A, X), reaching H:

$$D^2 = 3E^2 - A^2 - X, \quad I^2 = 2E^2 - A^2, \quad C^2 = 2E^2 - X,$$

$$F^2 = 2E^2 - D^2, \quad B^2 = 3E^2 - A^2 - C^2,$$

$$H^2 = 3E^2 - X - I^2 = 529, \qquad H = 23.$$

(D, X) reaches the same six values after $A^2 = 3E^2 - D^2 - X$ (column 1).

Column 3: (C,F)**,** (C,I)**,** (F,I)**.** (C,F) is Section 12. Represented instead by (C,I), reaching H:

$$F^2 = 3E^2 - C^2 - I^2, \quad A^2 = 2E^2 - I^2, \quad X = 2E^2 - C^2,$$

$$D^2 = 2E^2 - F^2, \quad B^2 = 3E^2 - A^2 - C^2,$$

$$H^2 = 3E^2 - X - I^2 = 529, \qquad H = 23.$$

(F,I) reaches the same six values after $C^2 = 3E^2 - F^2 - I^2$ (column 3).

Remark 13 (The shallow alternative). Every one of these twelve deep walks is optional. Once any single step produces X (equivalently C^2), the remaining five cells follow at once, in parallel, directly from (1)–(6) applied to A^2, X, and E^2 — two steps in total from any seed pair, not six. The six-step walks above are deliberately the longer route, chosen because each of their steps is individually a clean geometric identity, tracing the square rather than shortcutting through it. As with the chain (II1)–(II6) of Section 10, the number of layers records a choice of path, not a property of the cell reached; a fuller account of what the shallow path says about X itself is taken up in a later section.

15 The Anchor Basis and the Parity of the Elimination Chain

Three separate constructions — the elimination chain (II1)–(II6) of Section 10, the deep walks of Sections 11–14, and the shallow alternative of Section 14 — converge on the same destination, and doing so is not a coincidence but a consequence of rank.

Proposition 14. *Every relation expressing one cell as E^2 plus or minus exactly two other cells — a* triple *relation — uses E^2 with coefficient* 1*, never* $2E^2$*.*

Proof. The four anchor forms of Section 4,

$$B^2 = -E^2 + I^2 + X, \qquad D^2 = E^2 + I^2 - X, \qquad F^2 = E^2 - I^2 + X, \qquad H^2 = E^2 + A^2 - X,$$

already exhibit this. Any further triple relation is a rearrangement of one of these or of (1)–(6): for instance $A^2 = E^2 + C^2 - D^2$ (from column 1 combined with the corner pair) and $A^2 = E^2 + X - B^2$ (equation (3) rearranged) both hold, and both carry E^2 at coefficient 1. A relation with $2E^2$ instead would require two independent applications of Corollary 2 within a single triple, which needs four outer cells, not two, to balance. □

Remark 15. Among the four triples above, X is not merely present in some: it is present in *all four*, alternating sign $(+X, -X, +X, -X)$ as the cell cycles through B, D, F, H. The partner term alternates too, but in a different pattern — I^2 for B, D, F, since these are exactly the three vertices of the medial triangle around the anchors $\{I^2, X\}$ (the Medial configuration of Section 4), and A^2 for H, which sits outside that triangle as B's opposite-pair partner rather than a medial vertex. No triple relation among B, D, F, H omits X; each omits I^2 or A^2 instead. This is the precise sense in which X is unavoidable for these four cells specifically, and it is independent of, though consistent with, its role as the shared floor of Proposition 16.

Proposition 16. *The constants* $11, 9, 7, 5, 3, 1$ *of (II1)–(II6) are all odd, and no relation in that chain, extended in either direction, can close at an even multiple of* E^2*.*

Proof. Rearranged, (II6) reads $B^2 = -E^2 + I^2 + X$, which is exactly the anchor form above: coefficient 1, odd. Each subsequent step of the chain, by Section 10, trades one cell for its opposite-pair partner via Corollary 2, changing the constant by exactly $2E^2$. Starting from an odd coefficient and moving by even steps preserves parity indefinitely, so every constant reachable from (II6) — in particular the observed $1, 3, 5, 7, 9, 11$ — is odd. $12E^2$, or any even multiple, is therefore not merely absent from the six relations recorded: it is structurally unreachable by this chain, regardless of how far the elimination is extended. □

The anchor forms are the natural terminus of this descent because $\{E^2, I^2, X\}$ (Section 4) is already a minimal three-parameter basis: no further substitution can reduce a triple relation past coefficient 1, since doing so would require eliminating a fourth free quantity from a system that only has three. This is the same basis the shallow alternative of Section 14 reaches after two steps from *any* seed pair, and the same pair $\{X, A^2\}$ (or $\{X, I^2\}$) that dominates the recurrence counts of Section 13. The elimination chain descends toward it from above by removing pairs two at a time; the deep walks approach it from below by adding lines and pairs one at a time; the shallow alternative reaches it directly. All three are different routes to the identical resting point.

Remark 17. This sharpens, without overclaiming, the informal claim that X "closes" the square. X is not closing in every literal sense — it is itself a valid seed in four of the twelve pairs of Section 14, where it costs nothing to reference — but it is never more than one step from a seed pair that does not include it, and together with A^2 (or I^2) it is the fixed basis that every elimination, deep or shallow, is forced back toward. The elimination chain's confinement to odd multiples of E^2 is the precise, provable form of that fact: not that X is privileged over every other cell, but that the basis $\{E^2, I^2, X\}$ it belongs to is the unique floor beneath which no further reduction, by any route, is possible.

16 X as the Closing Cell

The term *closing cell* was introduced in Section 8 in a narrow, search-specific sense: once A and C are chosen so that A^2 and C^2 are square, X and B^2 are the two entries left over, fixed by the linear structure regardless of whether they themselves are square. Three independent lines of evidence, developed since, show that X closes the square in a second, purely algebraic sense as well — not tied to any particular search, and not a property of which two cells happen to be chosen as seeds.

In the chains. Section 13 compared two independently seeded deep chains and found that X, once it must be derived rather than handed over as a seed, recurs at least as often as any other derived quantity — outright in one chain, tied with A^2 in the other, never surpassed. Section 14 then showed that this is not particular to those two chains: of the twelve valid seed pairs, every deep walk passes through X at some point, and the shallow alternative available to all twelve reaches X (or already has it, when X is itself a seed) within a single step, after which every remaining cell follows in parallel.

In the E^2 relations. Proposition 14 and the remark following it show that among the four triples B^2, D^2, F^2, H^2 — the minimal, single-E^2 expressions for the four edge cells — X appears in *all four*, alternating sign, while its partner term alternates between I^2 and A^2. No minimal relation among these four cells omits X; each omits one of the two corner anchors instead.

In (II1)–(II6). X is not only the subject of the entire elimination chain of Section 10, appearing on both sides of (II1); Proposition 16 shows the chain's confinement to odd multiples of E^2 is inherited directly from X's presence in the anchor form at (II6), the chain's terminus. Every relation the chain passes through on its way from $11E^2$ down to E^2 is, in effect, a restatement of where X sits relative to the other eight cells.

Evidence	What it shows
Chains (13, 14)	X recurs at least as often as any derived quantity, across every seed pair tested, and is at most one step away when not itself a seed.
Triples (15)	X appears in all four minimal edge-cell relations; none omits it.
(II1)–(II6) (10, 15)	The chain is built on X throughout, and its odd-only parity is forced by X's role in the anchor form at the chain's end.

A single argument, from degrees of freedom alone, accounts for all three lines of evidence at once.

Proposition 18. *Fix A and its opposite-pair partner I (so $A^2 + I^2 = 2E^2$). No cell among B^2, C^2, D^2, F^2, H^2 is determined by A^2, I^2, and E^2 alone; each requires X as well.*

Proof. Since $A^2 + I^2 = 2E^2$, the pair (A^2, I^2) carries only one free number beyond E^2, not two. Proposition 4 requires exactly three free parameters to determine the family. One of the three is E² itself; a second is spent on A^2 (with I^2 fixed by it); a third remains unspent, and no relation among (1)–(6) can produce a fifth cell from only two free numbers. X is exactly the missing third parameter — indeed the same X of Proposition 4's own basis $\{A^2, E^2, X\}$ — and every one of B^2, C^2, D^2, F^2, H^2 requires it:

$$B^2 = -E^2 + I^2 + X, \quad D^2 = E^2 + I^2 - X, \quad F^2 = E^2 - I^2 + X,$$
$$H^2 = E^2 + A^2 - X, \qquad C^2 = 2E^2 - X.$$

□

Unlike the triples of Proposition 14, this is not restricted to B, D, F, H: it is a statement about the whole square, and it includes C^2 directly through its own simplest relation rather than through any longer detour. The apparent tension with Proposition 14's remark — that $A^2 = E^2 + C^2 - D^2$ is a valid X-free triple — is resolved by what that triple actually uses: D^2, a cell outside the closed vocabulary $\{A^2, I^2, E^2, X\}$ of Proposition 18. Within that vocabulary, X is not optional for any of the five remaining cells; escaping it requires importing a sixth quantity from outside, which is a different and more expensive kind of route, not a shorter one.

Remark 19. This is the precise and provable form of the informal claim that X closes the square starting from any single seed A: fix A, and its opposite pair I follows for free, but nothing else does without X. None of this makes X more important than A^2 in an absolute sense — Section 13 already established that A^2 ties X's recurrence when it, too, must be derived, and $\{A^2, E^2, X\}$ is a basis of three, not one. What Proposition 18 shows is narrower and more precise: of the three free parameters the family needs, one is always spent on E^2 and the natural second is spent on the pair (A, I) — and whichever cell is asked to supply the third, within that closed vocabulary it can only be X. That is the algebraic content behind calling X a closing cell, alongside its original, narrower meaning from Section 8.

Remark 20. Stated at its simplest: fixing E leaves exactly two degrees of freedom. Choosing the seed A spends the first, and fixes I with it at no extra cost, via $A^2+I^2 = 2E^2$. That leaves exactly one degree of freedom unspent, and B^2, C^2, D^2, F^2, H^2 — all five of the remaining cells — are reached only by spending it on X, each through nothing more than $\pm E^2$ and $\pm X$ together with the already-known A^2 or I^2. There is no other way to spend the second degree of freedom that reaches any of the five; X is not one option among several for closing the square in this framework, it is the only one, and it is the one term shared by all five closing equations alike.

17 The Sign Signature $(+1, -1)$ and Adjacency to X

With E fixed and the seed A spending the first degree of freedom (fixing I for free, Proposition 18), the second and only remaining degree of freedom is X. The sign each cell's coefficient-1 relation carries on E^2 and X is not arbitrary: it identifies precisely how the cell sits relative to X on the grid, and whether that placement is forced or merely one choice among two.

Proposition 21 (The X-sign rule)**.** *A cell's minimal relation carries $-X$ if it shares a row, column, or the corner opposite pair with X directly; it carries $+X$ if it is instead the opposite-pair partner of such a cell.*

Proof. D shares column 1 with X ($A^2+D^2+X = 3E^2$); H shares row 3 with X ($X+H^2+I^2 = 3E^2$); C is X's own opposite pair (Corollary 2). Solving each of these three identities for the cell in question produces $-X$ directly. B and F share no line with X and are not paired with it; they are instead the opposite-pair partners of H and D respectively, and $\text{partner}^2 = 2E^2 - \text{cell}^2$ negates every term of the partner's own equation, turning H's $-X$ into B's $+X$, and D's $-X$ into F's $+X$. ☐

Proposition 22 (Forced vs. free sign)**.** *Among the coefficient-*1 *triples for B^2, D^2, F^2, H^2: D^2 and H^2 admit exactly one sign assignment of E^2, together with I^2 or A^2 respectively; B^2 and F^2 admit two, one using A^2 and one using I^2, differing in the sign of E^2.*

Proof. Exhaustive search over $s_1E^2 + s_2\{A^2 \text{ or } I^2\} + s_3X$, $s_i \in \{+1, -1\}$, against the known numerical values finds exactly one solution for D^2, using I^2, and none using A^2; exactly one for H^2, using A^2, and none using I^2; and, for each of B^2, F^2, one solution using A^2 and a second, distinct, using I^2:

$$D^2 = E^2 + I^2 - X, \qquad H^2 = E^2 + A^2 - X,$$

$$B^2 = E^2 - A^2 + X = -E^2 + I^2 + X, \qquad F^2 = -E^2 + A^2 + X = E^2 - I^2 + X.$$

□

Combined, this identifies a genuine geometric signature. D, H, C — the three cells directly adjacent to X, by line or by pair — are forced into the single sign pattern $(E^2, X) = (+1, -1)$ (for C, doubled: $C^2 = E^2 + E^2 - X$), with no alternative available. B, F — reached only by stepping through an opposite-pair flip — are not forced: each admits two valid coefficient-1 forms, $(+1, +1)$ using its own natural anchor or $(-1, +1)$ using the translated one, and it is exactly this freedom that marks them as one step removed from X rather than directly adjacent to it.

Cell	Adjacent to X?	(E^2, X) sign	Forced?
D	column 1	$(+1, -1)$	yes, unique
H	row 3	$(+1, -1)$	yes, unique
C	opposite pair	$(+1, -1)$	yes, unique
B	no (partner of H)	$(+1, +1)$ or $(-1, +1)$	no, two forms
F	no (partner of D)	$(+1, +1)$ or $(-1, +1)$	no, two forms

Remark 23. The sign pattern is therefore diagnostic, not incidental: given any coefficient-1 triple for one of the five remaining cells, whether its (E^2, X) signs are forced to $(+1, -1)$ or free to be either $(+1, +1)$ or $(-1, +1)$ reveals, without consulting the grid at all, whether that cell sits directly on a line or pair with X or one step away from it. Searching for the signature $(+1, -1)$ is, in this precise sense, searching for the specific geometric formation of direct adjacency to X — and Proposition 21 guarantees the search succeeds for exactly three of the five, never more and never fewer.

18 Unique Formation for D, H, C

Fixing A, E, and X determines every cell of the square uniquely — Proposition 1 allows no other outcome. It is tempting to extend that uniqueness to the sign formation each cell is written in, but Proposition 22 already shows this is false for two of the five remaining cells, and the distinction is worth keeping sharp rather than rounding away.

- **Values are always unique.** For any A, E, X, each of B^2, C^2, D^2, F^2, H^2 takes exactly one value, with no ambiguity.
- **Formations are unique only for D, H, C.** Each of these three has exactly one valid sign template at coefficient 1 (Proposition 22): once A, E, X are fixed, no second formation matches.
- **Formations are two-way for B, F.** Each of these two has *two* valid sign templates — one per corner anchor — and both correctly reach the same value. Neither formation is privileged over the other; both match simultaneously.

So it is not correct to say that fixing A, E, X creates a single formation that no other can match, in full generality: that statement is exactly right for D, H, C, and exactly wrong for B, F, whose defining feature is precisely that two formations tie. The asymmetry is the result, not an exception to it: it is what separates the three cells directly adjacent to X (Proposition 21) from the two reached only through an opposite-pair flip, and collapsing it into a single blanket claim of uniqueness would erase the distinction Section 17 was built to establish.

19 The (C, E, X) Reflection and the Three-Panel View

The mirrored placements in the sign panels are not incidental: they are the trace of a single symmetry of the square, the reflection across the diagonal through C, E, X.

Proposition 24. *Reflecting the square across the diagonal C, E, X fixes C, E, and X, and swaps $A \leftrightarrow I$, $B \leftrightarrow F$, and $D \leftrightarrow H$.*

Proof. This is one of the eight symmetries of Section 26 ("Symmetries of the square"), the reflection through the anti-diagonal C, E, X rather than the main diagonal A, E, I used there as the illustrative rotation. Written as $(i, j) \mapsto (4 - j, 4 - i)$ on the grid coordinates, it fixes exactly the three cells on that diagonal and exchanges each remaining cell with the one occupying the mirrored position: A (top-left) with I (bottom-right), B (top-middle) with F (middle-right), D (middle-left) with H (bottom-middle). $\square$

Applying the swap $A^2 \leftrightarrow I^2$ together with $B \leftrightarrow F$ to the sign forms of Section 17 carries one exactly onto the other:

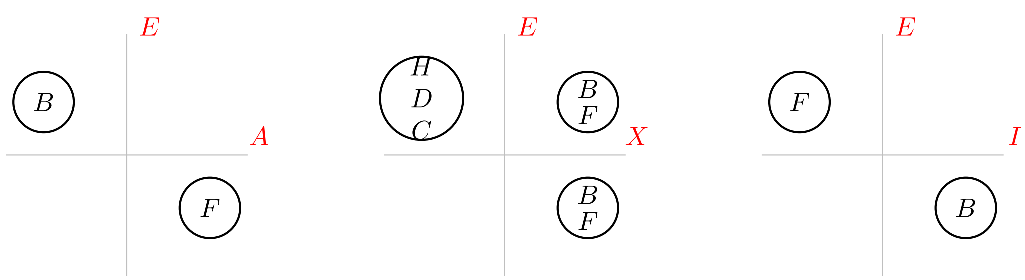
$$B^2 = E^2 - A^2 + X \xrightarrow{A^2 \leftrightarrow I^2,\ B \leftrightarrow F} F^2 = E^2 - I^2 + X,$$

$$B^2 = -E^2 + I^2 + X \xrightarrow{A^2 \leftrightarrow I^2,\ B \leftrightarrow F} F^2 = -E^2 + A^2 + X,$$

and the same swap, applied to $D \leftrightarrow H$, carries D's single forced form onto H's:

$$D^2 = E^2 + I^2 - X \xrightarrow{A^2 \leftrightarrow I^2,\ D \leftrightarrow H} H^2 = E^2 + A^2 - X.$$

The three panels below display exactly this: each plots two of A^2, I^2, X against E^2, and the reflection of Proposition 24 is the map that carries every label to its mirror image across the origin.



Remark 25. The reflection accounts for the mirroring visible across all three panels — it is the single symmetry that carries B's two forms onto F's two forms, and D's one form onto H's one form, all at once — but it does not by itself explain why B, F have two forms while D, H, C have only one. That asymmetry is fixed by the reflection, not produced by it: C and X lie on the mirror line, so their direct adjacency to X (Proposition 21) is preserved rather than exchanged, while $A \leftrightarrow I$ and $B \leftrightarrow F$ are genuinely swapped. The unique formation of D, H, C and the two-way formation of B, F (Section 18) are each individually preserved by this symmetry; the reflection relates the two halves of the free pair to each other, but the reason one pair is free and the other is not remains the adjacency argument, established independently of any rotation or reflection of the square.

20 Unique Formation for the Seed A and Scale E

Proposition 24 showed that the full reflection across C, E, X preserves every line of the square. It is worth checking directly that no smaller, partial rearrangement does the same — so that the reflection is not merely *a* symmetry that happens to work, but the only available move once A and E are fixed.

Proposition 26. *Once A and E are fixed, no partial swap of cells — exchanging a single opposite pair in isolation, or exchanging two cells that are not an opposite pair — preserves all eight lines of the square. Only the full reflection of Proposition 24, moving A, I, B, F, D, H together, does.*

Proof. Consider the square of Section 8. Swapping the opposite pair $B^2 \leftrightarrow H^2$ alone leaves column 2 (their shared line) correct, since it contains both and addition does not see their order:

$$B^2 + E^2 + H^2 = H^2 + E^2 + B^2.$$

But B also lies in row 1 and H also lies in row 3, and neither line contains the other cell of the pair:

$$A^2 + H^2 + C^2 = 181683 \neq 3E^2, \qquad X + B^2 + I^2 = 902067 \neq 3E^2.$$

The identical argument applies to the opposite pair $D^2 \leftrightarrow F^2$, via column 1 and column 3, and to the bare exchange $B^2 \leftrightarrow F^2$ (not an opposite pair at all), which fails row 1 and row 2 immediately. In every partial case, the swapped cells' *shared* line survives by commutativity, while at least one *other* line, containing only one member of the swap, does not. The full reflection avoids this because it moves every affected cell at once: substituting $A \leftrightarrow I$, $B \leftrightarrow F$, $D \leftrightarrow H$ simultaneously into all eight lines verifies each one directly. □

Swap attempted	Shared line	Other lines	Valid?
$B^2 \leftrightarrow H^2$ alone	column 2: holds	row 1, row 3: fail	no
$D^2 \leftrightarrow F^2$ alone	row 2: holds	column 1, column 3: fail	no
$B^2 \leftrightarrow F^2$ alone	(not paired)	row 1, row 2: fail	no
full (C, E, X) reflection	—	all eight: hold	yes

Remark 27 (The last-digit square). Since every odd square ends in 1, 5, or 9 (Section 17), and since reducing an identity modulo 10 preserves it, the eight line-sums of the square survive intact when every cell is replaced by its last digit. For the square of Section 8:

5	1	9
9	5	1
1	9	5

Every row, column, and diagonal — including the degenerate diagonal $5+5+5$ — sums to 15, matching $3E^2 = 541875 \equiv 5 \pmod{10}$. This is not a coincidence restricted to the full values: it is the same identity, viewed mod 10, and consequently the same rigidity applies. Each partial swap of Proposition 26 that breaks a real line breaks the corresponding digit line identically — exchanging B, H leaves the shared digit column correct but unbalances the two digit rows containing A, C and X, I — so the last-digit square is exactly as resistant to partial rearrangement as the square of squares itself, for the same structural reason.

Remark 28 (Constant digit grids have two causes). The digit square of Section 8's example varies across $\{1, 5, 9\}$, but a *constant* last-digit grid — every cell ending in the same digit — is also possible, and it is important not to read this as always signalling the trivial constant square of Section 10. It arises in two structurally different ways.

- **Degenerate:** every cell equals the same value t (Section 10), so every last digit is trivially identical. Here the digit square is constant because the square itself is constant.
- **Non-degenerate:** the six clean cells of the $E = 1105$ example, $A = 455$, $D = 1445$, $F = 595$, $H = 245$, $I = 1495$, $E = 1105$, are six *distinct* numbers, yet each ends in 5, because each is divisible by 5 — inherited from $1105 = 5 \cdot 13 \cdot 17$ — and any odd multiple of 5 ends in 5, not 0. Squaring preserves this, so every one of these genuinely different cells lands on the same last digit without any two of them being equal.

A constant digit grid is therefore not, by itself, evidence of the degenerate solution: it can equally well reflect a shared prime factor inherited from E. The reliable signature of degeneracy is constant *value*, not constant *digit*; the two coincide in the first case and diverge in the second.

Remark 29. This is the sense in which fixing the seed A and the scale E produces a *unique* formation, at the level of the square as a whole rather than any single cell: the nine values are pinned down completely (Proposition 1), and the only rearrangement of them that respects every line simultaneously is the one full reflection of Proposition 24 — not because smaller moves were not tried, but because each one necessarily leaves some line, real or reduced to its last digit, unbalanced. The square built from A and E is rigid: it admits exactly the eight symmetries of Section 26, of which this reflection is one, and nothing smaller than a full symmetry of the square preserves it.

21 Three Invariants Under Scaling

Fixing E, the seed A, and X pins down a single square. Rescaling E replaces that square by another, and it is natural to ask which parts of the formation built in Sections 16–20 survive the change and which do not. Three things happen, and they are not all the same kind of fact.

1. The two degrees of freedom. Proposition 18 holds for every E unconditionally: one degree of freedom is spent on the seed A (fixing I with it), and the second, with no alternative, on X. Rescaling E does not add or remove a degree of freedom — it is still exactly two, for every choice of centre.

2. The signs are scale-invariant. Rescale roots by k ($A, D, E \mapsto kA, kD, kE$) and X by k^2. This preserves (1)–(6) termwise, since each is linear and homogeneous in A^2, E^2, X:

$$B^2 = -E^2 + I^2 + X \implies k^2B^2 = -(kE)^2 + (kI)^2 + k^2X,$$

with the signs on the right untouched. Consequently Propositions 21 and 22 — which of B, C, D, F, H is forced into $(+1, -1)$ and which is free — hold identically after rescaling. This is confirmed directly: the representations of $2E^2$ at $E = 850 = 2 \cdot 425$ are exactly the representations at $E = 425$, doubled,

$$(7, 601), (85, 595), (175, 575), (205, 565), (289, 527), (329, 503), (355, 485) \longmapsto$$

$$(14, 1202), (170, 1190), (350, 1150), (410, 1130), (578, 1054), (658, 1006), (710, 970),$$

with no new pair appearing, since 850 introduces only a factor of 2 and no new prime $p \equiv 1 \pmod 4$ (Section 26). The same seed choice, doubled, therefore reproduces the identical sign formation on a larger scale.

3. The last-digit formation is not preserved, but its shape is. The specific digits $\{1, 5, 9\}$ of Section 20's digit square are a fact about $E = 425$, not about the formation itself, and rescaling changes them: at $E = 850$ ($k = 2$, so every value scales by $k^2 = 4$), the digit square becomes

0	4	6
6	0	4
4	6	0

— verified directly from $A = 410, D = 1054, E = 850$ — with every line still summing to the same constant, $3E^2 = 2\,167\,500 \equiv 0 \pmod{10}$. The digits $\{1, 5, 9\}$ became $\{0, 4, 6\}$, each the original multiplied by $k^2 = 4$ modulo 10, but the *arrangement* — which position holds which of the three digits, and the consequent balance of every line — is exactly the arrangement of Section 20, unchanged.

Remark 30. The pattern across all three is the same: what is invariant under scaling is the *relational* content — the count of degrees of freedom, which cells are forced and which are free, and which positions in the grid carry matching digits — while the *specific numbers* filling that relational structure are not. A formation, in this precise sense, is the shape common to a square and every rescaling of it; $E = 425$ and $E = 850$ are two instances of the same formation, not two different ones.

One invariant, not three. Section 5 already isolated the quantity that scale and translation leave untouched: placing F^2 at 0 and B^2 at 1,

$$t = \frac{D^2 - F^2}{B^2 - F^2}.$$

For the square of Section 8, $S = B^2 - F^2 = 277200$, and

$$t = \frac{194208}{277200} = 0.7006060606\ldots$$

De-normalising — scaling by S and translating by F^2 — reproduces every one of $F^2, E^2, X, D^2, I^2, B^2$ exactly. The square of $E = 425$ is therefore not a self-contained object: it is one scale-and-translation instance of the single point $t \approx 0.7006$ already identified in Section 5. Repeating the computation at $E = 850 = 2 \cdot 425$ gives

$$S = 1\,108\,800 = 4 \times 277200, \qquad t = 0.7006060606\ldots$$

again, exactly: the scale quadruples with k^2, and t does not move at all.

A cleaner scale for t. Reduced to lowest terms, $t = 578/825$, with $825 = 33 \times 25$. Fixing $F^2 = 0$ and $B^2 = 1$ and scaling by this denominator makes t itself land on an integer:

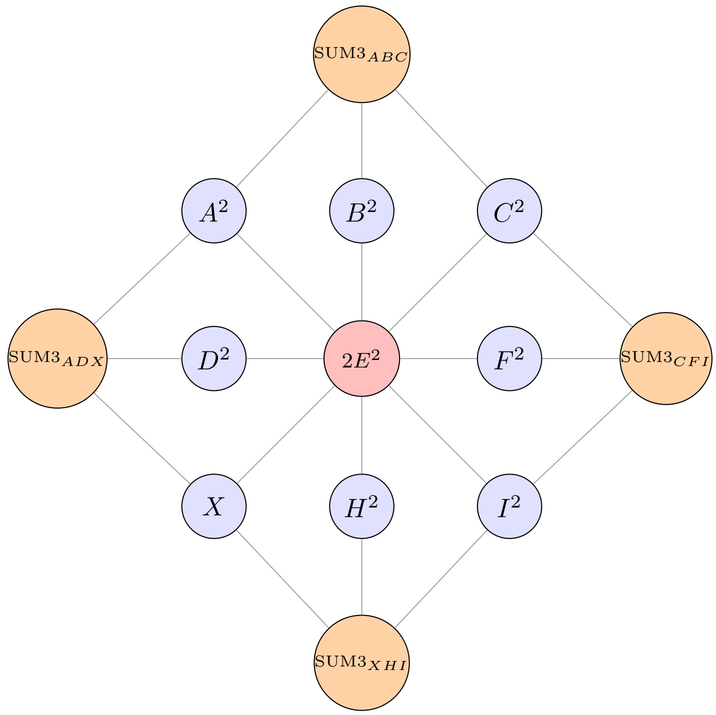
$$D^2 = 825\,t = 578, \qquad E^2 = \frac{825\,t}{2} = 289,$$

alongside $F^2 = 0$ and $B^2 = 825$ — four of the six anchors landing on whole numbers at this particular scale.

This identifies the single invariant behind all three results above. t is constructed to be exactly the scale-and-translation-invariant content of the family; the fixed count of two degrees of freedom, the scale-invariant sign formation, and the shape-preserved digit arrangement are three different windows onto the same fact, each visible in a different part of the structure — counting parameters, tracking algebraic signs, and reducing values modulo 10, respectively — but all three are already implied by, and give no more information than, the single number t that Section 5 isolated first.

22 The Incidence Graph

The linear system may be displayed as a weighted graph. The eight outer cells occupy their positions in the square; the centre is replaced by the node $2E^2$, into which every opposite pair feeds; and the four line-sums $3E^2$ sit outside, each joined to the three cells of its line. All edges carry weight 1.



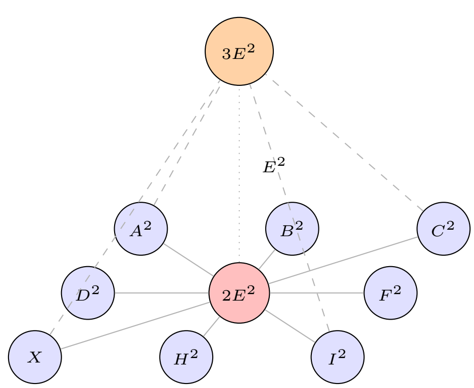
Each opposite pair — (A^2, I^2), (B^2, H^2), (C^2, X), (D^2, F^2) — lies diametrically across the hub and sums to $2E^2$. Adjoining the centre E^2 to any such pair yields a line, so the four outer nodes take the common value

$$\mathrm{SUM3}_{ABC} = \mathrm{SUM3}_{ADX} = \mathrm{SUM3}_{CFI} = \mathrm{SUM3}_{XHI} = 3E^2.$$

The graph therefore encodes the whole system: the four pair relations generate it, and the agreement of the four outer nodes is automatic.

23 Three-Dimensional Form

Placing the eight cells and the hub $2E^2$ in a base plane at height 0, and lifting each line-sum to height $3E^2$, gives a pyramid whose apex projects onto the centre.



Every line of the square consists of the centre together with one opposite pair, so each of the four ascending edges rises from a pair summing to $2E^2$ through the centre E^2 to the common apex $3E^2$. The apex projects vertically onto E^2, and the height of the pyramid is the magic sum.

24 The Cone $u^2 + v^2 = 2e^2$

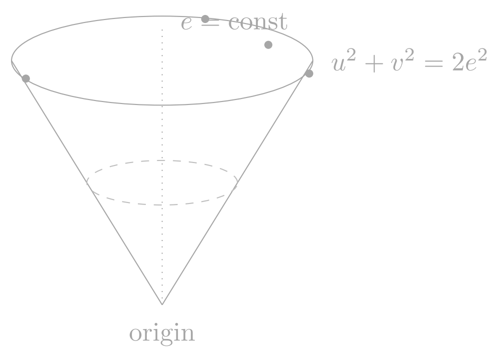
Write the entries of a nine-square magic square as $a^2, b^2, \dots, i^2$ with $X = g^2$. By Corollary 2 the four opposite pairs satisfy

$$a^2 + i^2 = b^2 + h^2 = c^2 + g^2 = d^2 + f^2 = 2e^2.$$

Each is an instance of the single equation

$$u^2 + v^2 = 2e^2,$$

which in (u, v, e)-space is a homogeneous quadratic, hence a circular cone with vertex at the origin and axis the line $u = v = e$. The plane $e = \text{const}$ meets it in the circle $u^2 + v^2 = 2e^2$ of radius $e\sqrt{2}$, on which the point (e, e) always lies. This point is the degenerate solution $u = v = e$, corresponding to the constant square.



Proposition 31. *A 3×3 magic square of nine perfect squares exists if and only if there is an integer e and four points*

$$(a, i),\ (b, h),\ (c, g),\ (d, f)$$

with integer coordinates on the circle $u^2 + v^2 = 2e^2$, pairwise distinct as unordered pairs and none equal to (e, e), satisfying the single additional linear condition

$$a^2 + b^2 + c^2 = 3e^2.$$

Proof. Corollary 2 gives the four circle conditions, and the top row gives the linear one; conversely, given such data, the parametrisation of Section 2 reconstructs the square, and distinctness of the pairs is exactly distinctness of the entries. □

Thus the entire problem reduces to the arithmetic of one number. Integer points on $u^2 + v^2 = 2e^2$ correspond to representations of $2e^2$ as a sum of two squares, whose number is governed by the primes $p \equiv 1 \pmod 4$ dividing e; a candidate centre must therefore be divisible by at least four such primes, or by prime powers supplying the equivalent count. Every solution of $u^2 + v^2 = 2e^2$ arises, up to scaling, as

$$u = m^2 + 2mn - n^2, \qquad v = |m^2 - 2mn - n^2|, \qquad e = m^2 + n^2,$$

so the four pairs are four choices of (m, n) with common e.

Remark 32. The contrast with Theorem 8 is the point of this section. The linear structure of the square is completely understood and cannot obstruct a solution; the obstruction, if there is one, lies in the impossibility of choosing four points on a single circle $u^2 + v^2 = 2e^2$ subject to one linear relation. This is a question about representations by sums of two squares, and it is where the difficulty of the problem resides. Intersecting the circle conditions with the linear condition yields a curve of genus one, which is the route by which elliptic curves enter the subject.

25 The Circle for $e = 425$

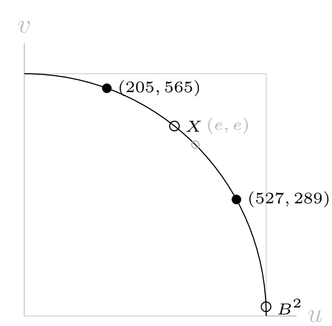
Fix $e = 425$. The four opposite pairs of the square of Section 8 give four points on the circle

$$u^2 + v^2 = 2e^2 = 361250,$$

of radius $425\sqrt{2} \approx 601.04$. Two of these points have integer coordinates and two do not:

$$(205, 565), \quad (527, 289), \quad \big(\sqrt{360721},\, 23\big), \quad \big(373,\, \sqrt{222121}\big).$$

The last two are the closing cells B^2 and X, whose entries are the non-squares 360721 and 222121.

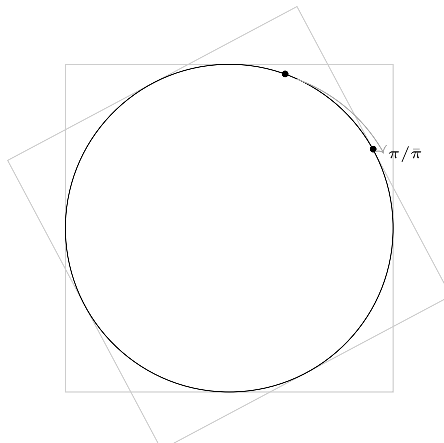


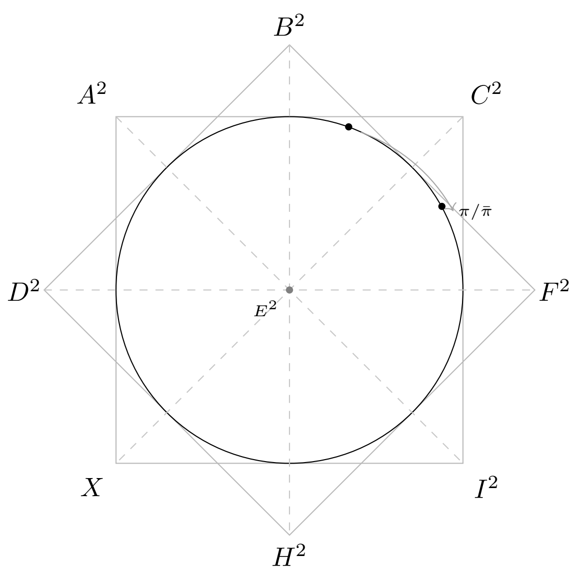
All four points lie exactly on the circle: the pair identity $u^2 + v^2 = 2e^2$ holds for every 3×3 magic square, whether or not its entries are squares. What distinguishes the closing cells is not their position but their coordinates — filled points are integral, open points are not.

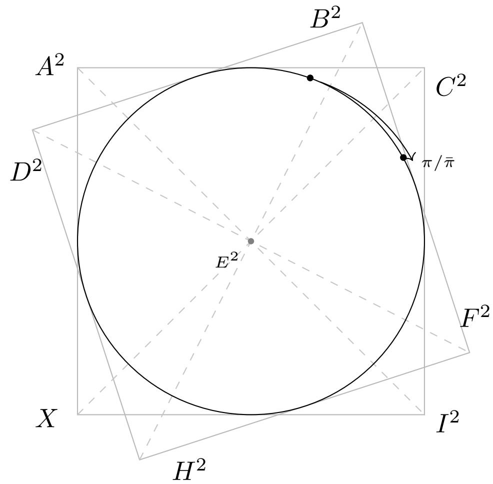
Remark 33. The circle is inscribed in the square $[0, e\sqrt{2}]^2$, and it is tempting to read the region between them as the failure of squareness. This is not correct. Every magic square places its four pairs *on* the circle; the distinction is between integral and merely real points of the same curve, and the integral points do not form a region. A nine-square magic square requires all four points to be lattice points, which is why the obstruction cannot be exhibited geometrically and must be settled arithmetically.

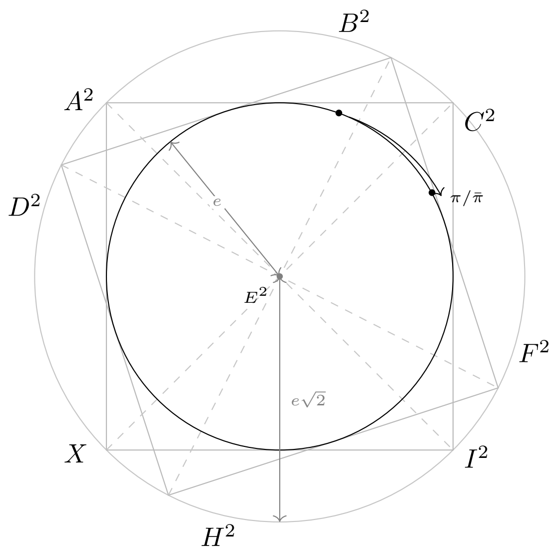
26 Rotation and the Gaussian Integers

The rotational picture of the previous section has an exact arithmetic meaning. Identify the point (u, v) on the circle $u^2 + v^2 = 2e^2$ with the Gaussian integer $u + vi$, whose norm is $2e^2$. Multiplication by a Gaussian integer of norm 1 — that is, by ± 1 or $\pm i$ — permutes the point among the eight symmetries of the configuration, while multiplication by a ratio $\pi/\bar{\pi}$, where π is a prime of norm $p \equiv 1 \pmod 4$ dividing e, rotates it to a *different* representation of the same norm.









Thus the representations of $2e^2$ as a sum of two squares form a single orbit under this rotation group, whose size is determined by the factorisation of e: if $e = 2^\alpha \prod p_j^{\beta_j} \prod q_k^{\gamma_k}$ with $p_j \equiv 1$ and $q_k \equiv 3 \pmod 4$, the number of essentially distinct representations grows with $\prod(2\beta_j + 1)$. A nine-square magic square requires at least four of them, and hence a centre divisible by several primes $p \equiv 1 \pmod 4$.

The three-dimensional form of the same statement is a rotation of the pyramid of Section 8 about its axis: the base circle is carried to itself, the apex is fixed, and the four ascending edges are permuted among the available representations. The configuration is therefore invariant under the rotation, and no rotation can separate the circle from its circumscribed square.

Remark 34. This closes the geometric account. The rotation group acts transitively on the representations, but which representations exist is a property of the integer e and not of the figure. Consequently neither the planar nor the three-dimensional picture can decide the problem: they exhibit the orbit, while the question is whether the orbit is large enough, and suitably positioned, to satisfy the linear condition $a^2 + b^2 + c^2 = 3e^2$. That question is arithmetic.

27 Two Kinds of Rotation

The rotating-square figure admits two distinct motions, which should not be conflated.

Symmetries of the square

Rotation through a multiple of 90°, together with the four reflections, generates the dihedral group D_4 of order 8. In the Gaussian picture these are multiplication by the units $\pm 1, \pm i$ and conjugation, sending

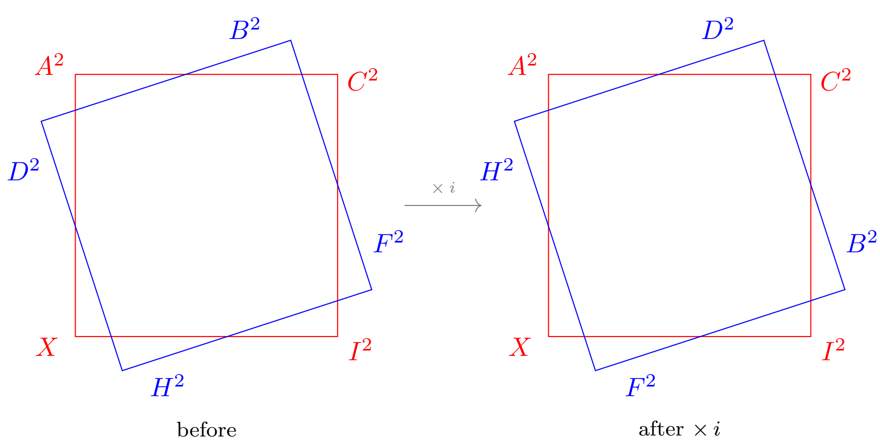
$$(u, v) \longmapsto (-v, u), \quad (-u, -v), \quad (v, -u), \quad (v, u),$$

each of which preserves $u^2 + v^2$. Applied to a magic square, such a motion permutes the cells — carrying B^2 to the position formerly held by F^2, and so on — while leaving the multiset of nine entries unchanged. The result is the same square in a different orientation, not a new solution. Every magic square therefore occurs in an orbit of eight, and solutions are counted up to this action.

Rotation between representations

The second motion is arithmetic. If π is a Gaussian prime of norm $p \equiv 1 \pmod 4$ dividing e, then multiplication by $\pi/\bar{\pi}$, an element of norm 1 but not a unit, carries a lattice point of $u^2 + v^2 = 2e^2$ to a *different* lattice point of the same circle. The angle of rotation is $2 \arg \pi$, which is not in general a multiple of 90° and depends on p. For $e = 425 = 5^2 \cdot 17$, this action carries $(205, 565)$ to $(527, 289)$, exhibiting two of the four representations used by the square of Section 8.

Remark 35. The distinction is essential. Rotations of the first kind are available for every e and produce nothing new; rotations of the second kind produce genuinely distinct representations, but only in numbers governed by the factorisation of e. A nine-square magic square requires four representations related by rotations of the second kind, subject to $a^2 + b^2 + c^2 = 3e^2$. The figure displays the orbit; whether a suitable quadruple exists within it is not a geometric question.



28 Search Design and the Closing Cells

The reduction of Section 24 suggests an obvious search: fix e, enumerate the integer points of $u^2 + v^2 = 2e^2$, and select two of them as the pairs (a, i) and (c, g), so that A^2, I^2, C^2, X and E^2 are square by construction. The remaining cells B^2, H^2, D^2, F^2 are then determined, and one tests how many of them are squares.

Each design answers a different question, and it is worth recording which. Drawing c from the list of representations forces

$$X = 2e^2 - C^2$$

to be a perfect square, so the search ranges over configurations in which both pairs (A^2, I^2) and (C^2, X) are square. This is the appropriate range for the nine-square question, where every cell must be square in any case. It is a narrower range than the study of intermediate configurations requires: in the square of Section 8, $X = 222121$ lies strictly between 471^2 and 472^2, its non-square entries being precisely the closing cells B^2 and X, so that configuration falls outside this range. At $e = 425$ the constrained search accordingly returns only configurations with the five square entries it guarantees by construction.

Relaxing the constraint does not enlarge the supply of squares; it admits configurations in which the non-square entries fall on the closing cells. In both designs C^2 ranges over squares. What differs is whether its partner $X = 2e^2 - C^2$ is required to be square as well. Draw a from the representations of $2e^2$, fixing the pair (A^2, I^2); then let C^2 range over all squares less than $2e^2$, with X free to be a non-square. The five remaining cells follow from (1)–(6), and the number of square entries is counted rather than imposed.

29 A Density Heuristic for the Closing Cells

Suppose the pair (C^2, X) is broken, so that X is not a square. Since

$$B^2 - X = E^2 - A^2,$$

the value of B^2 differs from X by a fixed amount, and is therefore also a non-square except by coincidence. The count of square entries then turns on the single remaining cell

$$H^2 = 2E^2 - B^2 = E^2 + A^2 - X,$$

which is square in a seven-square configuration and not in a six-square one.

The magnitude of H^2 is governed by A relative to E. Writing $A = \alpha E$ with $\alpha > 0$, we have $H^2 = (1 + \alpha^2)E^2 - X$, so H^2 is small when $\alpha < 1$ and large when $\alpha > 1$, for X in the admissible range. Since consecutive squares near n differ by approximately $2\sqrt{n}$, the proportion of integers near n that are squares decreases like $n^{-1/2}$. A small forced value of H^2 therefore has a higher chance of being square than a large one.

This gives a heuristic account of the two configurations at $e = 425$. With $A = 205 < E$, the forced value is $H^2 = 529 = 23^2$, and the square attains seven square entries. With $A = 527 > E$, the same construction forces $H^2 = 236233$, which lies between 486^2 and 487^2, and the count falls to six. The known seven-square example accordingly has $A < E$.

Remark 36. The argument is heuristic and not a proof. It gives a reason to expect seven-square configurations to be found more readily when $A < E$, but it excludes nothing: H^2 is a specific integer, not a random one, and large squares occur freely elsewhere in the same square — for instance $F^2 = 565^2 = 319225$. Establishing that no seven-square configuration exists with $A > E$ at a given e would require an exhaustive check over the admissible values of C, which the density argument does not supply.

30 Admissibility and the Choice of A

Fix E and C, so that $X = 2E^2 - C^2$ is determined, and write

$$\delta = X - E^2.$$

The parametrisation then gives

$$H^2 = A^2 - \delta, \qquad F^2 = A^2 + \delta,$$

so H^2 is positive precisely when $A > \sqrt{\delta}$. This is a genuine restriction: roots below the threshold cannot serve as A at all, independently of any squareness condition.

Among the admissible roots, $H^2 = A^2 - \delta$ is increasing in A. The smallest admissible root therefore yields the smallest forced value of H^2. Since consecutive squares near n differ by roughly $2\sqrt{n}$, the density of squares near n falls like $n^{-1/2}$, and a smaller forced value has a correspondingly better chance of being square. This gives a heuristic rule:

> Among the roots available at a given centre, the most promising choice of A is the smallest one exceeding $\sqrt{\delta}$.

At $E = 425$ with $C = 373$ we have $X = 222121$ and $\delta = 41496$, so $\sqrt{\delta} \approx 203.7$. The available roots are $23, 205, 289, 373, 527, 565$; the root 23 falls below the threshold and is inadmissible, and 205 is the smallest root above it. It yields $H^2 = 529 = 23^2$, and the resulting square attains seven square entries. The next admissible root, 289, yields $H^2 = 42025 = 205^2$ — still square, but the configuration achieves only six, the D/F pair failing instead.

Remark 37. Two cautions. First, the rule is heuristic: H^2 is a determinate integer, and its squareness is not a matter of probability. The example above shows the rule identifying the best configuration while the decisive failure at $A = 289$ occurs elsewhere in the square, at the pair (D^2, F^2). Second, replacing A by its partner $\sqrt{2E^2 - A^2}$ is the reflection across the anti-diagonal, which permutes the entries without altering their multiset; the count is therefore invariant under that exchange, and one may normalise to $A < E$ without loss.

31 An Admissibility Threshold

Proposition 38. *Fix E and C, and set $X = 2E^2 - C^2$ and $\delta = X - E^2$. Then*

$$H^2 = A^2 - \delta, \qquad F^2 = A^2 + \delta,$$

and the nine entries are all positive only if

$$A > \sqrt{\delta} \quad \textit{when } \delta > 0, \qquad A > \sqrt{-\delta} \cdot 0 \quad \textit{(no lower constraint)} \quad \textit{when } \delta \leq 0.$$

Proof. From (4), $H^2 = E^2 + A^2 - X = A^2 - \delta$, and from (6), $F^2 = A^2 + X - E^2 = A^2 + \delta$. If $\delta > 0$ then $H^2 > 0$ requires $A^2 > \delta$; if $\delta \leq 0$ then $H^2 \geq A^2 > 0$ automatically, while $F^2 > 0$ requires $A^2 > -\delta$. In either case the smaller of the two cells imposes the bound $A^2 > |\delta|$. □

Corollary 39. *Roots A with $A^2 \leq |\delta|$ are inadmissible: no magic square with positive entries has that value in the corner cell, whatever the squareness of the remaining cells.*

The threshold is a strict consequence of the parametrisation, not a heuristic. It applies before any question of squareness arises, and it removes candidates from the search unconditionally.

At $E = 425$ with $C = 373$ one has $X = 222121$ and $\delta = 41496$, whence $\sqrt{\delta} \approx 203.7$. Of the roots $23, 205, 289, 373, 527, 565$ occurring at this centre, the root 23 lies below the threshold and is inadmissible; the remaining five are admissible. Taking $A = 205$, the smallest admissible root, gives $H^2 = 42025 - 41496 = 529 = 23^2$, and the configuration attains seven square entries.

Remark 40. Since $H^2 = A^2 - \delta$ increases with A, the smallest admissible root produces the smallest forced value of H^2. As the density of squares near n decreases like $n^{-1/2}$, this suggests examining the smallest admissible root first. That suggestion is heuristic; Proposition 41 itself is not.

32 An Admissibility Threshold

Proposition 41. *Fix E and C, and set $X = 2E^2 - C^2$ and $\delta = X - E^2$. Then*

$$H^2 = A^2 - \delta, \qquad F^2 = A^2 + \delta,$$

and all nine entries are positive if and only if

$$A^2 > |\delta| = |X - E^2|.$$

Proof. From (4), $H^2 = E^2 + A^2 - X = A^2 - \delta$; from (6), $F^2 = A^2 + X - E^2 = A^2 + \delta$. Their minimum is $A^2 - |\delta|$, so both are positive exactly when $A^2 > |\delta|$. Equality gives a vanishing entry, which is excluded. The remaining cells are positive whenever $C^2 < 2E^2$ and $A^2 < 2E^2$, which the choice of C and A already assumes. □

Corollary 42. *Any A with $A^2 \leq |X - E^2|$ is inadmissible: no magic square with positive entries carries that value in the corner cell.*

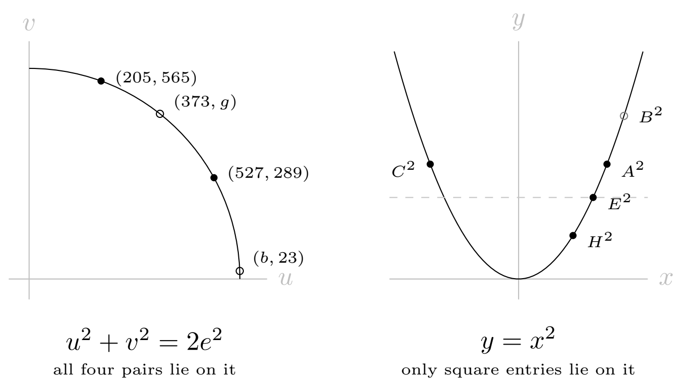
The bound is a consequence of the parametrisation alone. Its derivation uses only the linear identities (4) and (6), and makes no assumption about whether X, A^2, or any other entry is a perfect square. It therefore applies to every configuration — including those, such as the square of Section 8, in which X is not a square — and may be used to prune a search before any squareness test is performed.

At $E = 425$ with $C = 373$ one has $X = 222121$, which is not a square, and $\delta = 41496$, so $|\delta|^{1/2} \approx 203.7$. Of the roots $23, 205, 289, 373, 527, 565$ occurring at this centre, only 23 falls below the threshold. Taking $A = 205$, the smallest admissible root, gives $H^2 = 42025 - 41496 = 529 = 23^2$ and a configuration with seven square entries.

Remark 43. Since $H^2 = A^2 - \delta$ increases with A when $\delta > 0$, the smallest admissible root produces the smallest forced value of H^2; as the density of squares near n decreases like $n^{-1/2}$, this suggests testing that root first. The suggestion is heuristic. Proposition 41 is not.

33 The Two Conditions, Side by Side

The two conditions defining the problem are of different kinds and live in different coordinate planes. The left panel shows the pair condition $u^2+v^2 = 2e^2$, satisfied by every magic square; the right panel shows the squareness condition $y = x^2$, on which only some of the entries lie.

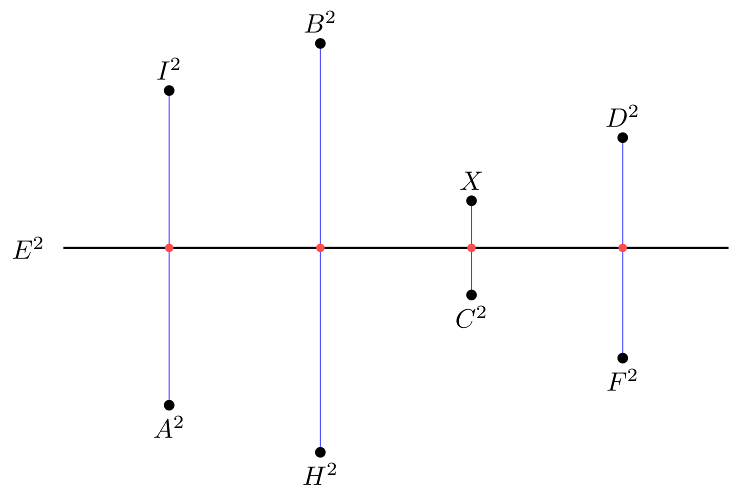


In the left panel all four pairs of any magic square lie exactly on the circle, since $u^2+v^2 = 2e^2$ is an identity of the family; two of the four have integer coordinates and two, marked hollow, do not. In the right panel an entry n appears as the point $(\sqrt{n}, n)$, which is a lattice point precisely when n is a perfect square; the entry B^2, marked hollow, is not.

Remark 44. The two panels cannot be superimposed. The axes of the left are two roots u, v; the axes of the right are a root and its square. A scaling of the circle therefore cannot carry it onto the parabola, and no single figure displays both conditions at once. This separation is the geometric counterpart of Theorem 8: the pair condition is uniform across the family and visible, while the squareness condition is arithmetic and is not.

34 The Pairs as Bisected Segments

Plotting each entry at its own value and drawing the horizontal line $y = E^2$ gives the magic condition in its simplest form: every opposite pair is a vertical segment bisected by that line.



The four segments are drawn to scale for the square of Section 8. Each is bisected by $y = E^2$, since $A^2 + I^2 = B^2 + H^2 = C^2 + X = D^2 + F^2 = 2E^2$; the marked midpoints all coincide in height with the line. The segments differ greatly in length — from 82992 for the pair (C^2, X) to 360192 for (B^2, H^2) — and their lengths are the only freedom the magic condition leaves, the three independent half-lengths being the three parameters of the family.

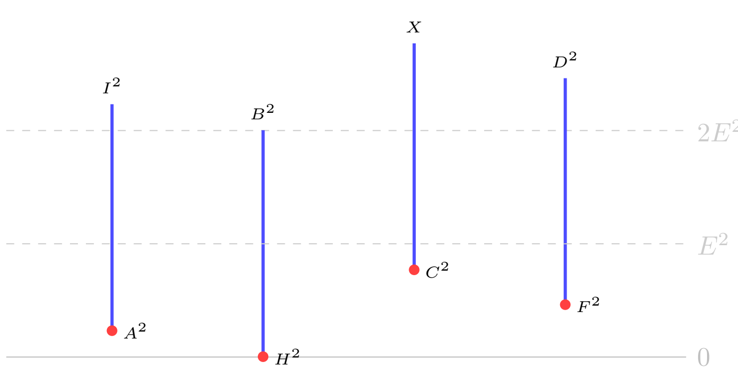
Remark 45. The figure displays the magic condition completely and says nothing about squareness: it is identical in form for every 3×3 magic square. The entries $B^2 = 360721$ and $X = 222121$ are not squares, yet their segments are bisected exactly as the others are.

35 The Pairs as Segments of Equal Length

Each opposite pair may be drawn as a vertical segment whose lower endpoint is the smaller entry and whose upper endpoint is the larger, both measured from a common baseline at 0. Since

$$A^2 + I^2 = B^2 + H^2 = C^2 + X = D^2 + F^2 = 2E^2,$$

the difference between the endpoints is not constant, but the *sum* is; drawing each segment so that its length equals that sum renders all four segments congruent, of common length $2E^2$. The four pairs then differ only in elevation.



The figure is drawn to scale for the square of Section 8, with heights expressed as fractions of $2E^2 = 361250$. The bases sit at

$$\tfrac{H^2}{2E^2} = 0.0015, \quad \tfrac{A^2}{2E^2} = 0.1163, \quad \tfrac{F^2}{2E^2} = 0.2312, \quad \tfrac{C^2}{2E^2} = 0.3851,$$

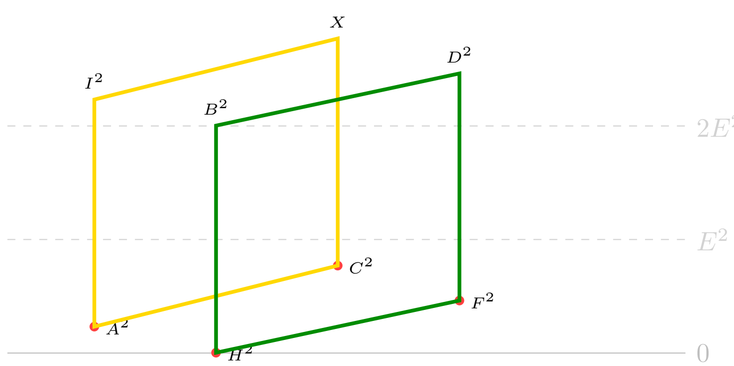
and each summit follows at exactly one unit above.

Three features are worth noting. The congruence of the segments is the magic condition itself, expressed without arithmetic: it is what $u + v = 2E^2$ looks like when u and v are drawn as the ends of a single interval. The elevation of each segment is unconstrained, and the three independent elevations are precisely the three parameters of the family, the fourth being determined once the others are chosen. Finally, a segment based at height E^2 would have both endpoints equal to E^2; the distance of each base below that level therefore measures how far the corresponding pair is from the degenerate configuration.

Remark 46. As with the preceding figures, the picture is indifferent to squareness. The segments for (C^2, X) and (B^2, H^2) are congruent to the others and equally well placed, although $X = 222121$ and $B^2 = 360721$ are not perfect squares.

36 The Two Quadrilaterals

Joining the four corner cells A^2, I^2, X, C^2 and, separately, the four edge cells H^2, B^2, D^2, F^2, each set forms a parallelogram: the two segments in each set are congruent, of length $2E^2$, so opposite sides are parallel by construction.



The yellow quadrilateral carries the corner cells and the green the edge cells; they correspond respectively to the upright and the rotated square of the figure in Section 26, in which the same two sets appear as the vertices of two squares inscribed in the circle $u^2 + v^2 = 2E^2$. Since each vertical side has length $2E^2$, both quadrilaterals are parallelograms of equal side length, differing only in the elevations of their two sides.

Remark 47. The correspondence between the two figures is exact: a pair of cells is a vertical side here and a diameter there, and in both cases the defining relation is $u + v = 2E^2$. Neither figure registers whether an individual entry is a perfect square.

37 The Partner Function and Automatic Pair Generation

The identity $u^2 + v^2 = 2E^2$ of Corollary 2 may be turned into a function that manufactures partners on demand, converting the search of Section 24 into a short finite computation.

Definition 48. *For $w \geq 0$ define*

$$f(w) = \Big(\sqrt{w}\,(\sqrt{w}+1) - \sqrt{w}\Big)^{1/2}, \qquad g(x) = f\big(2E^2 - x\big), \qquad x \leq 2E^2.$$

Although f is written as a nested radical, it collapses algebraically:

$$\sqrt{w}\,(\sqrt{w}+1) - \sqrt{w} = w + \sqrt{w} - \sqrt{w} = w,$$

so $f(w) = \sqrt{w}$ identically, and hence $g(x) = \sqrt{2E^2 - x}$. The nested form is kept here only because it is the one in which the function was first written down; numerically the two forms agree, but the nested form accumulates more floating-point error, performing three chained square roots where one suffices.

Proposition 49. *For every $x \leq 2E^2$, writing $v = g(x)$ gives $x + v^2 = 2E^2$ identically.*

Proof. $v^2 = g(x)^2 = 2E^2 - x$ by Definition 48. □

Proposition 49 is the mechanism, not a discovery: it guarantees the pair relation for *any* input x, whether or not v is itself an integer. To search for pairs of *cells* — both members squares of integers — apply g to a squared root rather than to a raw value:

$$q(x) = g(x^2) = \sqrt{2E^2 - x^2}, \qquad r(x) = q(x)^2 = 2E^2 - x^2.$$

By Proposition 49, $x^2 + q(x)^2 = 2E^2$ holds for every integer x; what varies is only whether $q(x)$ is itself an integer.

Proposition 50 (Automatic pair generation)**.** *For given E, the set*

$$\mathcal{P}(E) = \big\{\,(x,\, q(x)) : 1 \leq x \leq E,\ q(x) \in \mathbb{Z}\,\big\}$$

is exactly the set of representations of $2E^2$ as a sum of two positive squares (Section 26), together with the degenerate pair (E,E). Every element of $\mathcal{P}(E)$ may be assigned to any one of the three symmetric opposite-pair roles (A,I), (D,F), (H,B), since each satisfies the same identity $u^2 + v^2 = 2E^2$ underlying all three, and none of the three formulas (1)–(6) *distinguishes which role a given representation fills.*

This turns the search into a scan: for $x = 1, \ldots, \lfloor E\sqrt{2} \rfloor$, retain those x with $r(x)$ a perfect square, and read off the pair $(x, q(x))$. At $E = 425$ the scan returns exactly the seven pairs

$$(7,601),\ (85,595),\ (175,575),\ (205,565),\ (289,527),\ (329,503),\ (355,485),$$

together with the degenerate $(425,425)$. Assigning $(A,I) = (205,565)$ and $(F,D) = (289,527)$ — two of the seven pairs, chosen with no further arithmetic — reproduces the square of Section 8 completely once run through (1)–(6): $C = 373$, $H = 23$, $X = 222121$, and $B = 360721$ follow immediately. Five of the nine cells, E, A, I, F, D, are guaranteed square by this construction alone, before any further search; the remaining question — whether C, X, H, B also cooperate — is exactly the content of the sections that follow.

38 Closing the Third Pair in Closed Form

Section 37 showed that drawing two of the three symmetric pairs directly from $\mathcal{P}(E)$ — say (A, I) and (D, F) — guarantees five squares for free. It is natural to ask whether a *third* pair, (C, X), can be tested just as cheaply, without scanning a probe x against a lookup table. It can: the first column of the square supplies a second identity that, combined with Corollary 2, pins C down in one step.

Proposition 51. *For any positive integers A, D, set $X = 3E^2 - A^2 - D^2$, the value forced by the first-column condition $A^2 + D^2 + X = 3E^2$. Then*

$$C^2 = A^2 + D^2 - E^2$$

identically, so C is an integer precisely when $A^2 + D^2 - E^2$ is a non-negative perfect square — with no search over X, and no table of the values $r(x)$, required.

Proof. By Corollary 2, $C^2 + X = 2E^2$, so $C^2 = 2E^2 - X$. Substituting $X = 3E^2 - A^2 - D^2$ gives $C^2 = 2E^2 - (3E^2 - A^2 - D^2) = A^2 + D^2 - E^2$. □

Proposition 51 is the closed form of the check that the table $\{r(x)\}$ was performing implicitly: testing whether the column's forced X equals $r(x) = 2E^2 - x^2$ for some x is the same test as asking whether $2E^2 - X$ — namely $A^2 + D^2 - E^2$ — is itself a perfect square, and the latter needs no table at all. What was an $O(E)$ membership test against a precomputed set collapses to a single `isqrt` check per candidate pair (A, D).

At $E = 425$, applying Proposition 51 to every ordered pair drawn from $\mathcal{P}(425)$ recovers $C^2 = 205^2 + 527^2 - 425^2 = 139129 = 373^2$ at once — the square of Section 8 — with no scan. The same closed form also flags two further admissible, distinct-cell configurations, $(A, D) = (527, 85)$ and $(527, 205)$, each reaching six squares (E, A, I, D, F, C) but not seven: swapping which member of a pair is labelled A rather than D changes the forced $H^2 = E^2 + A^2 - X$, and in these two cases H^2 fails to be square where, for $(A, D) = (205, 527)$, it happens to succeed. This sharpens the moral of Section 40: reaching seven squares is not a property of *which* representations are used, but of the specific role-assignment among them, and Proposition 51 makes that assignment cheap enough to check exhaustively.

Remark 52. Writing $S = A^2 + D^2$, Proposition 51 and the corner identity $C^2 + X = 2E^2$ together read as two complementary measurements of the same quantity:

$$C^2 = S - E^2, \qquad X = 3E^2 - S.$$

Adding them cancels S entirely, $C^2 + X = (S - E^2) + (3E^2 - S) = 2E^2$, recovering Corollary 2 for this pair regardless of what S is. C^2 measures how far $A^2 + D^2$ sits above the centre value E^2; X measures how far it falls short of the magic sum $3E^2$; and the two measurements are complementary simply because E^2 and $3E^2$ are themselves symmetric about $2E^2$.

Remark 53. This complementarity is purely algebraic and carries no arithmetic force: it says nothing about whether C and $\sqrt{X}$ are integers, only that whichever real values they take, their squares sum correctly. Squareness of C (Proposition 51) and squareness of X are two independent conditions on the same derived quantity C^2 — the first, that $S - E^2$ is a perfect square; the second, that C is itself a member of $\mathcal{P}(E)$, i.e. that $2E^2 - C^2$ is *also* a perfect square. Requiring both is a strictly rarer event than requiring either alone: an exhaustive check over every ordered triple drawn from $\mathcal{P}(E)$ finds no (A, D, C), at either $E = 425$ or $E = 1105$, with C satisfying Proposition 51 *and* lying in $\mathcal{P}(E)$ simultaneously. Beyond the guaranteed five (E, A, I, D, F) and the sixth cell C of Proposition 51, every configuration found in Sections 38–40 that reaches a seventh square gets it from H, never from X — consistent with X requiring this second, independent, and so far unmet condition.

The independence is not an artefact of small examples: at $E = 5 \cdot 13 \cdot 17 \cdot 29 = 32045$, where $\mathcal{P}(E)$ already contains forty pairs, the same exhaustive triple search still returns no coincidence. Nor is there a parity obstruction of the kind found in Section 40 to explain the absence — odd squares are $\equiv 1 \pmod 8$ regardless of which odd number is squared, so $A^2 + D^2 - E^2 \equiv 1 \pmod 8$ matches C^2 automatically for every choice of odd A, D, C, E, and no modulus rules the coincidence out. The scarcity is instead the single-cell shadow of the remark closing Section 24: Proposition 51 alone is one conic condition on C, easily satisfied and easily parametrised, which is why $\mathcal{P}(E)$ itself is cheap to generate; requiring C to *also* lie in $\mathcal{P}(E)$ imposes a second, independent conic condition on the same variable, and two conics intersected generically cut out a curve of genus one rather than genus zero — sparse where each condition alone was rich. The difficulty of pushing a single cell from six squares to seven by this route is therefore the same difficulty, in miniature, that Section 24 identifies as the obstruction to a nine-square solution overall: not a wall of divisibility, but a drop from a curve with infinitely many easily-found points to one with only finitely many, possibly none.

39 The Unsimplified Form, and a Name for the Outcome

Proposition 51 presents C^2 in its reduced form, $A^2 + D^2 - E^2$. It is instructive to write it unreduced, exactly as it is derived, since doing so displays both identities that produce it before they combine.

Combining the corner identity $C^2 + X = 2E^2$ (Corollary 2) with the column identity $A^2 + D^2 + X = 3E^2$, and solving for C directly rather than eliminating X algebraically first, gives

$$C = \sqrt{2E^2 - \left(3E^2 - A^2 - D^2\right)}\,.$$

At $E = 425$, $A = 205$, $D = 527$, this evaluates numerically to

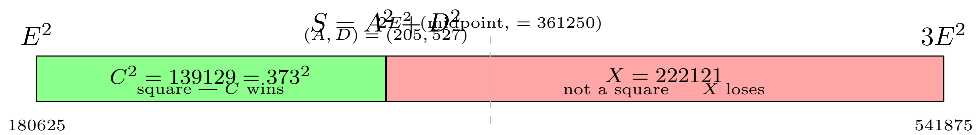
$$C = \sqrt{2 \cdot 425^2 - \left(3 \cdot 425^2 - 205^2 - 527^2\right)} = \sqrt{139129} = 373,$$

matching the square of Section 8. Only once the inner parenthesis is evaluated do the $2E^2$ and $3E^2$ cancel down to the single relation of Proposition 51; nothing of the two source identities survives as a separate, still-active constraint after the arithmetic is carried out — C answers to one condition, not two.

Remark 54. It is convenient, and harmless, to describe the outcome for a given (A, D) in the language of a contest: at $(A, D) = (205, 527)$, "C wins" the coincidence of Proposition 51

— $C = 373$ is an integer — while "X does not," since $X = 222121$ is not a perfect square. This is a name for the result, not a claim about its cause: C^2 and X are two numbers summing to $2E^2$, and nothing in their derivation makes either's squareness bear on the other's. The genuine reason X so rarely joins C — confirmed absent even among the forty representation pairs of $E = 32045$ in Section 38 — is the genus-one sparseness already identified in Section 24, not a contest between the two identities used to derive C, which are fully spent once C itself is computed.

The outcome is easiest to see drawn to scale, as a single bar running from E^2 to $3E^2$ — a span of length $2E^2$, with $2E^2$ itself sitting at the exact midpoint — split at $S = A^2 + D^2$ into the two segments C^2 and X.



Drawn to scale, the asymmetry is visible directly: S sits well below the $2E^2$ midpoint for this (A, D), so X's segment is the larger of the two, purely as a consequence of where S happens to fall — not of any contest between the segments themselves.

40 A Parity Obstruction, and the Role of $p \equiv 1 \pmod 4$

The search of Section 37 fixes E and one opposite pair — say A, with I its automatic partner — and then probes a second root, C, over a range of integers, checking whether the resulting X, F^2, H^2, D^2, B^2 happen to be perfect squares. We show that this probe may be restricted to odd C without loss, and relate the underlying mechanism to the classical condition on prime factors of E governing representations of $2E^2$.

Lemma 55. *If E is odd and u, v are positive integers with $u^2 + v^2 = 2E^2$, then u and v are both odd.*

Proof. Since E is odd, $E^2 \equiv 1 \pmod 4$ and $2E^2 \equiv 2 \pmod 4$. Squares are $\equiv 0$ or $1 \pmod 4$ according as their root is even or odd. If u, v have opposite parity, $u^2 + v^2 \equiv 0 + 1 = 1 \pmod 4$; if both are even, $u^2 + v^2 \equiv 0 \pmod 4$. Neither matches $2 \pmod 4$, so u, v must both be odd. □

Consequently, for odd E, every entry A that occurs as one member of a genuine integer pair (A, I), (D, F), or (H, B) is itself odd; this is why every value appearing in the representation lists of Section 24 (for instance $205, 565, 289, 527$ at $E = 425$, and all thirteen pairs at $E = 1105$) is odd.

Proposition 56. *Fix odd E and odd A with $A^2 + I^2 = 2E^2$ for some integer I, and let C be any positive integer with $C \neq A, E$, giving $X = 2E^2 - C^2$ and the forced cells of (1)–(6). If C is even, then X, F^2, and B^2 are each $\equiv 2 \pmod 4$, hence none of X, F, B can be a perfect square.*

Proof. With E, A odd, $E^2 \equiv A^2 \equiv 1 \pmod 4$, so $2E^2 \equiv 2 \pmod 4$. If C is even, $C^2 \equiv 0 \pmod 4$, so $X = 2E^2 - C^2 \equiv 2 \pmod 4$. Then

$$F^2 = A^2 + X - E^2 \equiv 1 + 2 - 1 = 2 \pmod 4, \qquad B^2 = E^2 - A^2 + X \equiv 1 - 1 + 2 = 2 \pmod 4.$$

Since no perfect square is $\equiv 2 \pmod 4$, neither F^2 nor B^2 can be square, and X itself, being $\equiv 2 \pmod 4$, cannot be square either. □

Corollary 57. *Under the hypotheses of Proposition 56, if C is even then at most six of the nine cells can be perfect squares: the four guaranteed cells E, A, I, C, together with at most H and D, since $H^2 = E^2 + A^2 - X \equiv 0 \pmod 4$ and $D^2 = 3E^2 - A^2 - X \equiv 0 \pmod 4$ remain admissible residues. In practice the ceiling is lower still: an exhaustive search at $E = 425$ and $E = 1105$ found no even C exceeding five squares in total.*

Odd C carries no such obstruction — all four of X, F^2, H^2, D^2, B^2 are admissible residues in that case — and it is exactly among the odd probes that the seven-square example of Section 8 is found: there, $A = 205$ and $C = 373$ are both odd, and three of the four remaining cells ($H^2 = 529$, $D^2 = 277729$, $F^2 = 83521$) land on perfect squares simultaneously, while only $X = 222121$ fails.

Remark 58. Proposition 56 halves the search: for odd E, only odd values of the probe C need be tested, since even C is provably incapable of reaching the closing cells X or B, and is empirically no better than the guaranteed baseline for H, D either. This is a genuine pruning rule, established unconditionally by parity, in the same spirit as Proposition 41's positivity bound.

We now turn to a coarser but prior question: whether a genuine (non-degenerate) integer pair A, I exists at all. By the classical theorem on sums of two squares, $2E^2$ admits a representation as a sum of two *distinct* positive squares if and only if E has at least one prime factor $p \equiv 1 \pmod 4$; the number of essentially distinct representations grows with the number and multiplicity of such primes, as recorded in Section 26. Three worked centres illustrate how loosely this richness controls the attainable square count.

E	factorisation	representations of $2E^2$	best square count found
425	$5^2 \cdot 17$	7	7
1105	$5 \cdot 13 \cdot 17$	13	6
422	$2 \cdot 211$	0	5

At $E = 422$, 211 is prime and $211 \equiv 3 \pmod 4$; since 211 divides E to an odd power and no prime $\equiv 1 \pmod 4$ divides E at all, $2E^2$ admits no non-trivial representation as a sum of two squares. No pair (A, I), (D, F), or (H, B) can therefore ever consist of two distinct positive integers, and the search collapses to isolated single squares chosen independently, capping the total well below what 425 or 1105 achieve despite 422 being a comparably sized centre.

Conversely, 1105 has *more* representations than 425 — three distinct primes $\equiv 1 \pmod 4$ against 425's two — yet an exhaustive search over every representation-pair and every odd probe C (Corollary 57) found no configuration exceeding six squares, short of 425's seven. Richness of representation is therefore necessary in the weak sense of Proposition 56 and its corollary — without it, as 422 shows, the ceiling drops sharply — but it is not sufficient: whether a *specific* pairing of two representations (or of one representation with an unrestricted odd probe) also satisfies the forced linear relations of (1)–(6) is a finer question that the count of representations alone does not answer.

41 Conclusion

The family of 3×3 magic squares is a three-parameter linear family. We have given its complete parametrisation (1)–(6), shown that the associated system has rank 6 and nullity 3, and exhibited a basis for its solution space. Every relation among the cells derivable from the magic conditions is a combination of these six, and holds identically on the family. It follows (Theorem 8) that no linear relation can be violated by requiring the entries to be perfect squares — which accounts for a persistent feature of algebraic approaches to the problem, namely that successive elimination yields correct relations closing at $11E^2$, $9E^2$, $7E^2$, $5E^2$, $3E^2$ and E^2, none of which excludes any configuration.

Recast in the anchors (E^2, I^2, X), three of the cells take the symmetric form (7)–(9), and the configuration is the medial triangle of B^2, D^2, F^2. Up to similarity the whole family is one-dimensional, parametrised by a single ratio t. The cells B^2 and X close the configuration: once the other seven are fixed, these two are determined, and whether they are squares is not settled by the linear structure.

The arithmetic content is isolated in Section 24: a nine-square magic square exists if and only if some integer e admits four essentially distinct representations of $2e^2$ as a sum of two squares, subject to the single linear condition $a^2 + b^2 + c^2 = 3e^2$. Equivalently, four lattice points on the circle $u^2 + v^2 = 2e^2$, related by the Gaussian rotations of Section 11 and constrained by one plane. Since the number of such representations is governed by the primes $p \equiv 1 \pmod 4$ dividing e, this is a question about representations by sums of two squares, not about the geometry of the square.

Two further results are unconditional. Proposition 41 gives the admissibility bound $A^2 > |X - E^2|$, which holds regardless of squareness and may be used to prune a search in advance. And the reflection exchanging A^2 with I^2 preserves the multiset of entries, so the number of square cells is invariant under it, permitting the normalisation $A < E$.

The maximum number of perfect squares known to occur in a 3×3 magic square remains seven, realised by the square of Bremner and Sallows, whose two non-square entries occupy the closing cells B^2 and X. Whether eight or nine are attainable is open. What the present analysis establishes is where the question does *not* reside: not in the linear relations among the cells, which are fully understood and uniformly satisfied, but in the arithmetic of a single integer $2e^2$ and the lattice points of the circle it defines.